DATA SYSTEMS

CloudPro

Point cloud georeferencing, cleaning, classification and delivery

STAGE 03 OF THE DCS CHAIN · POSPRO → ALIGNPRO → CLOUDPRO

User Guide

For CloudPro 2.2.2 · November 2026

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1 Introduction

CloudPro is stage 03 of the Canadian Lidar Data Collection Systems (DCS) chain. It turns the Lidar returns recorded by the mobile mapper in `.clraw` captures into a clean, classified and georeferenced point cloud, and delivers it for the job.

- **Georeferencing.** Every Lidar return is placed at its own firing time, with the trajectory PosPro finalized and the Lidar mounting, lever arm and latency AlignPro finalized. Without AlignPro the mounting recorded in the capture is used, and every result says *no calibration*.
- **Viewing.** A Plan view with the project grid and a rotatable Inspection Area, a Slice view for cross-sections and profiles, and an interactive 3D view, all showing the points of the Inspection Area.
- **Manual editing.** Reject, accept, select or classify points with a rectangle, polygon, lasso or brush, above or below a drawn line, inside or outside the shape.
- **Automatic filters.** 36 filters in groups, with a live preview before anything changes: cell statistics, clips, outliers, snow and rain, ground, height above ground, walls and reflections, moving objects, and the extraction of road markings, curbs, poles, traffic signs, power lines, manhole covers and guardrails.
- **Keeping up with the chain.** When PosPro or AlignPro finalize again, CloudPro georeferences the job again by itself as soon as everything upstream is current (after a new PosPro trajectory it waits for AlignPro's Apply, unless the job uses

the recorded or entered mounting), with the settings of the last georeferencing and every edit carried over. With CloudPro's window closed this runs in the background, with a progress bar and Cancel in the QGIS message bar; the job opens in CloudPro when it has finished. With the job open in CloudPro it runs there. Nothing is finalized automatically. Turn it off per job in the Georeference dialog (`UPDATE AUTOMATICALLY WHEN POSPRO OR ALIGNPRO FINALIZE`); the `GEOREFERENCE` step in the header still rebuilds the points on request.

- **Delivery.** Finalize writes a LAS 1.4 cloud, an optional ground grid, a report and `cloud.json` for the job. Export writes LAS, XYZ or grids anywhere.

The mapper's captures are never changed. Everything CloudPro makes is in the job's processing project, in `03_CloudPro/`.

A job in five steps

1. **Open the job's project** from the project box, or click `03 CLOUDPRO` in PosPro's or AlignPro's header. PosPro, and ideally AlignPro, should have finalized first.
2. **Georeference:** `PROJECT MENU › GEOREFERENCE / UPDATE THE POINTS…` (`Ctrl+I`). Check the trajectory and mounting the dialog shows, then press `GEOREFERENCE`.
3. **Look around:** move the Inspection Area in the Plan view; the Slice and 3D views follow it.
4. **Clean and classify:** in `AUTOMATIC FILTERS` choose a group, press `LIVE PREVIEW`, check the result, then `ACCEPT` or `RUN` it on the whole project. Correct the rest by hand.
5. **Finalize:** `PROJECT MENU › FINALIZE FOR THE JOB…` (`Ctrl+P`).

What is new

VERSION	CHANGES
2.2.2	Steady offsets: along the road and sideways from walls only, the along-road offset checked part by part (Steady offsets). On Shirley 2.2.1 reported 8.4 cm per pass along the road; most of it was a height difference read on the slope.
2.2.1	**Steady offsets:** each line pair's steady offset in the vehicle's frame (along, sideways, up, roll), reported as calibration (AlignPro) when it reverses with the driving direction or grows with the distance from the drive lines, else as trajectory (PosPro); problem areas are what is left after it, so one steady offset no longer makes hundreds of areas; findings written for AlignPro 1.8.0 (Steady offsets).
2.2.0	**Line separation:** how far overlapping lines sit apart, measured on flat surfaces (ground, roofs, walls) and shown on the Plan, per line pair and per line, with the shift that would line them up, and the local **problem areas** with the line at fault and the time, which PosPro 2.2.0 shows in its Trajectory Review. Nothing is changed (Line separation).
2.1.0	**Linked cursor:** the point under the mouse in one view is marked in the other two: a cross on it in the Plan and the 3D view, a dashed vertical line at its place in the Slice (Views and navigation).
2.0.3	Each view keeps its own colour ramp and point size (they changed every view); a linked view follows the ramp of the view it is linked to. The Plan, Slice and 3D titles are the same size as the panel titles, with each view's controls right after its title. The Classification legend shows the class names without their numbers. The small window showing *PosPro trajectory · AlignPro calibration* no longer pops up (that text is the Georeference step's tooltip).
2.0.2	Class colours without clashes: every class colour is clearly different from every other one and from the rejected (magenta), preview (orange, green) and selected (white) colours. Class 15 (transmission tower) is blue, road markings (68) pale cyan instead of white, and 0, 3, 5, 10, 11, 13, 19, 20, 65, 66, 69, 70, 71 and the unlisted classes changed slightly or more (table in Classes).
2.0.1	Class 7 (low point, noise) is dark red instead of magenta, so it is not mistaken for rejected points when a view is coloured by Classification.
2.0.0	**New layout:** the header holds the chain and CloudPro's steps (Georeference › Clean & classify › Finalize) as breadcrumbs, with the guidance in their tooltips; the project box holds CloudPro's menu and the edit tools sit right of it (no MENU button, no tool bar). Views and panels stay inside the window (× hides them). One legend per view, without title or counts: right-click it for the colours, link, grid and sizes. The Slice has a `WIDTH` list (0.5 to 16 m) and whole slices centred inside the Inspection Area, with first / last buttons and *n / N*. The Inspection Area is sized in whole metres. *Publish* is now *Finalize*. Needs DCS Core 1.11.0.
1.9.9	**Project-wide filters on every core:** runs on the entire project or selected lines split dense tiles into smaller parts and run the parts at the same time, as many as fit in memory; the progress line shows parts and points done, parts running and the time left; `CANCEL` writes nothing. The performance log records each run's parts, threads and memory. Number and name boxes follow the dark theme.
1.9.8	**Performance report:** CloudPro records how long each step takes (loading an Inspection Area, filters and previews, georeferencing, publishing, exporting, 3D frames and whether the graphics card draws them) in `03_CloudPro/log/performance.log`. `PERFORMANCE REPORT...` on the project menu saves it with a description of the computer as one text file to send for review (Large areas).
1.9.7	**3D view on the graphics card:** with a graphics card (OpenGL 2.1 or later) the points are copied to the card once and every frame draws all of them in one call, moving or still, so the 3D view no longer thins the cloud while you turn it. Eye-dome shading is computed by the engine on every core. Without a usable card (or with software OpenGL) the 3D view draws on the processor as before; the status bar says which is used.
1.9.6	**Background georeferencing:** when PosPro or AlignPro publish and the chain is current again, CloudPro georeferences the job by itself (in the background when its window is closed) with the last settings and every edit carried over, so the job opens ready to clean. Needs DCS Core 1.10.0.
1.9.5	**Faster views on big areas:** the Plan, Slice and 3D views are drawn by the CloudPro engine on every core in one pass over the points (the Plan about 5 to 7 times faster on one core at 5–25 million points, the 3D view about 4 times; the same picture at point size 1); loading an Inspection Area skips tiles outside it, takes tiles inside it whole and reads tiles in parallel. The status bar says where the loading time went (read, order, prepare, draw), and slow loads are noted in the History.

1.9.4	**Export thinning and tiles:** THIN TO a grid spacing from a list (with the point density shown, 3D or 2D cells, the real point nearest each cell centre kept) and TILES of 50 to 1000 m named by their corner (LAS and XYZ, also from the command line: `--spacing`, `--thin-2d`, `--tile`); the grid cell is a list too. **Much faster filters on dense data:** SNOW / RAIN / DUST (DROR, LIOR, DIOR) and RADIUS OUTLIER no longer slow down as the point density rises (5 to 12 times faster on a dense façade, more on denser data; exactly the same results). RUN stops a live preview that is still computing and reuses a finished one of the same group instead of computing it again; SOR works in blocks, so memory stays bounded. After a run, the filter panel and the History log give the time of each filter.
1.9.3	**Upgrades work while QGIS is open:** the CloudPro engine is loaded from a copy in your user folder, so Windows no longer locks the plugin folder (1.9.0–1.9.2 could make an upgrade fail with *Failed to remove the directory*).
1.9.2	**Clearer editing:** the selection shape is red for Reject, green for Accept, grey for Select; rejected points are magenta (no longer listed on the scale bars); the Inspection Area outline is white. Window tabs along the bottom; view titles without point counts; the Plan cell size is a list. In 3D, dragging (and the left / right arrows) turns the cloud the way you move.
1.9.1	**Easier navigation:** a ZOOM TO THE INSPECTION AREA button in every view and ZOOM TO THE EXTENTS in the Plan; the 3D view turns around the centre of the Inspection Area and follows it when it moves; the 3D navigation arrows turn 22.5° per click. **Colour range per view:** right-click the scale bar for automatic over the Inspection Area or the whole project, or a fixed minimum and maximum (Colours); linked views follow the view they are linked to.
1.9.0	**CloudPro engine:** georeferencing (packets, Hesai decoding, skipping stopped packets, georeferencing, tiles) runs in a compiled library on every core, about 3 times faster per core; the grids are built in parallel; MOVING OBJECTS, GUARDRAILS, POLES, TRAFFIC SIGNS and VERTICAL PLANE use it too. Exactly the same results as before (see Large areas).
1.8.1	**Faster filters and grids:** the morphological ground filter (SMRF) is about 8 times faster with exactly the same ground points; MOVING OBJECTS is quicker; the grids behind the views are built twice as fast and, after an edit, in the background, so the window no longer pauses.
1.8.0	**Dockable windows:** Plan, Slice, 3D, Automatic filters, Lines · Classes · Info and History are each a window that can be moved, resized, tabbed together, floated or closed; MENU › WINDOWS reopens them or resets the layout, and the layout is remembered. Closed or hidden views are not drawn. **Big jobs:** SKIP POINTS WHILE STOPPED when georeferencing, faster import, less memory per loaded point, faster refresh after edits, one UPDATE GEOREFERENCING prompt (on the workflow strip), and a check of the point store on every open. **Rejected points hidden by default**, with an eye per view to show them in red; views colour and draw only the points they show. **Faster 3D view:** about twice as fast per frame, and the full-detail picture after a move is drawn in the background so the window stays responsive.
1.7.1	**Coordinate epoch** in Project properties › Geodetics: the points are delivered at the provincial NAD83(CSRS) epoch (New Brunswick 1997.0 for New Brunswick Stereographic, else the federal 2010.0), or as observed. A trajectory in another epoch is moved with NRCan's velocity grid. Needs DCS Core 1.9.0.
1.7.0	**Update keeps your work safe:** the new points are built beside the old ones and only replace them when georeferencing succeeds, so a cancel, an error or a trajectory that does not cover a line no longer loses manual and filter edits. Actions that change points or the project wait while a job runs; closing the window stops the job. **PPP trajectories** from PosPro (ITRF2020 at the observation date) are moved to NAD83(CSRS)v8 before georeferencing (about 1.4 m horizontally and 1.1 m vertically in Atlantic Canada); `cloud.json` and the report say so. `cloud.json` takes its frame and epoch from the trajectory actually used. Exports are written under a temporary name and removed when cancelled. Faster grids after whole-project filters and a faster Poles filter. Map layers and menus fixed for QGIS 4. Needs DCS Core 1.8.0.
1.6.0	Guided workflow: a strip under the header shows the three steps (Georeference, Clean & classify, Publish) with their state, and one NEXT button does the next thing (Georeference, or Update when the inputs changed; the Automatic filters; Publish for the job). The version number is shown after the plugin name in the header. Needs DCS Core 1.6.0.
1.5.3	Fix: 1.5.2 from the plugin repository would not open (a module and this guide were missing from the download).
1.5.2	Field check before PPK: the Plan grid shows *Density* (pts/m²) and *Coverage* against a target density you set (green on target, amber from half, red below), with the share on target, the area covered and the median density; `cloudpro info` prints the same. Works on the Certus real-time track, before PosPro and AlignPro.
1.5.1	Mapper 2.10 projects: the survey lines are recognised from what the files record, so a line split over several files is one line and older captures grouped into `lines/` read too, in recording order; navigation recordings are left out. Needs DCS Core 1.4.0.

1.5.0	The same header as the other DCS plugins: logo, plugin name, workflow boxes beside it, then CloudPro's inputs line, **UPDATE** and **MENU**, the project box, the Window button and the user guide button. The window title is just *CloudPro*. This rewritten guide, with pictures of this version. The 3D navigation arrows are half the size. Needs DCS Core 1.3.0.
1.4.0	DCS Core's standard window: floating the first time, docked only from the Window menu, never by dragging; it comes back as you left it, also after QGIS restarts; **RESET WINDOW**. Neutral graphite look with lighter view backgrounds. Needs DCS Core 1.2.0.
1.3.0	`.clraw` reading, Lidar and INS decoding, SBET and LAS come from DCS Core (shared with AlignPro), as does the Zstandard decoder. A missing Zstandard decoder is reported instead of every chunk counting as damaged. Needs DCS Core 1.1.0.
1.2.x	The user guide inside the plugin (1.2.3). The 3D navigation arrows (1.2.2). Cell size of the Plan grid on the tool bar, corner arrow on cycling tool buttons, colour scale bar in every view, linked view colours, tuning of the filters to the sensor precision, point budgets, memory check, autosave (1.2.1). The four-quarter window, Project Properties, the icon tool bar, colour by file and the seven feature extraction filters with classes 67–71 (1.2.0).
1.1.x	Part of the DCS chain: the shared project, PosPro's trajectory and AlignPro's mounting, **UPDATE** keeping edits, publishing with `cloud.json`, relocation of a moved job folder, one log per run, editing lock per job.

2 Installing and opening CloudPro

CloudPro is one of three QGIS plugins that process a Canadian Lidar mobile mapper job one after the other. All three need the **DCS Core** plugin, which holds what they share: the job's processing project, the files each stage hands to the next, the look, the header and the window.

STAGE	PLUGIN	READS	FINALIZES
01	**PosPro**	The mapper captures (`.clraw` or `.anpp`), base station data	The trajectory: `trajectory.json` with SBET and SMRMSG
02	**AlignPro**	The captures and PosPro's trajectory	The Lidar mounting: `alignment.json`
03	**CloudPro**	The captures, the trajectory and the mounting	The point cloud: `cloud.json`, LAS 1.4, ground grid, report

Requirements

- QGIS 3.34 or later, including QGIS 4.
- DCS Core 1.11.0 or later. CloudPro 2.2.2 will not start with an older DCS Core: a message bar asks you to update it.

Install from the Canadian Lidar repository

1. In QGIS open `PLUGINS › MANAGE AND INSTALL PLUGINS › SETTINGS`, press **Add...** and enter the repository URL `https://canadian-lidar.github.io/qgis-plugins/plugins.xml`.
2. Press **Reload all Repositories**, then install **DCS Core** first and **CloudPro** after it.
3. Restart QGIS. Later versions appear under `UPGRADEABLE` in the same window; update DCS Core together with the plugins.

Without internet access, use `INSTALL FROM ZIP` in the same window with `dcs_core-1.11.0.zip` first, then the CloudPro zip.

Opening CloudPro

- `PLUGINS › CANADIAN LIDAR › DATA SYSTEMS › CLOUDPRO`, or the CloudPro button on the Plugins toolbar.
- From another DCS plugin: click the **CloudPro** box in its header to open CloudPro on the same project.
- This guide: the `?` button in the header, `USER GUIDE (THIS VERSION)` in the project box, or `HELP › PLUGINS › CLOUDPRO USER GUIDE`. The guide that opens always matches the installed version.

3 The CloudPro window

PosPro, AlignPro, CloudPro and RemixPro share the same header and the same kind of window, so what you learn here applies to all of them.



Figure 3.1 The header. Left: the logo, the plugin name and version, then two rows of breadcrumbs: the chain of stages and this plugin's own steps. Then the project box (which also holds CloudPro's menu), the edit tools, and the user guide button at the right.

The header

PART	WHAT IT DOES
Plugin name and version `PosPro v2.1.0`	The plugin and the version you are running. Quote it when you report a problem; the user guide (`?`) is the one for this version.
The chain `PosPro › AlignPro › CloudPro`	The state of each stage on the open project, by colour: **green** finalized and current, **amber** out of date (an earlier stage was run again, the raw data or a shared setting changed), **blue** running, **red** failed, **grey** not run yet. The plugin you are in is underlined. Hover a name for the state and the reasons; click another stage to open that plugin on the same project. The chain refreshes every few seconds.
CloudPro's steps `GEOREFERENCE › CLEAN & CLASSIFY › FINALIZE`	CloudPro's three steps, which are actions: **Georeference** (or update the points when the inputs changed; its tooltip says what the points were georeferenced with), **Clean & classify** (opens the Automatic filters; optional, as long as you like) and **Finalize** (finalize for the job). The next step to do is underlined; clicking a step does it. Each step is coloured the same way (green done, amber needs a check, red needs attention, blue running, grey not done yet); the current one is underlined. Hover a step for what is done and what to do next; click it to go there.
Project box	The open project's name; click it for the project menu (Projects); it also holds CloudPro's own menu: Project properties, Georeference, Finalize, Export, Save, the QGIS map tools, other captures, the memory limit, Remove all points, Windows, Keys and mouse (Layout, menu and tools). It turns red while CloudPro is working.
Edit tools	Pan, the selection shape, the line tools, the edit action, inside / outside, undo and redo, measure, zoom all views to the Inspection Area, zoom the Plan to the project, and the rejected points eye (Layout, menu and tools).
`?`	Opens this guide.

Moving through the steps

CloudPro is an editor rather than a list of pages: its steps in the header are the guide. Click `GEOREFERENCE`, `CLEAN & CLASSIFY` or `FINALIZE` to do that step; hover a step for its state.

The window

CloudPro opens as a **window of its own**, centred on the QGIS screen the first time. Use the system's title bar to move it, maximise, minimise or close it; it does not dock into the QGIS window. CloudPro remembers its place and size, or that it was maximised, and opens the same way next time, also after QGIS restarts. `RESET WINDOW` in the project menu brings it back to its default size, centred.

4 Projects and the job folder

PosPro, AlignPro and CloudPro work on one shared **processing project** per job. The mapper's capture folders are never changed: everything the plugins make goes in a `Processing` folder.

PATH	HOLDS
`Job_Oct20/`	The job folder: the capture folders recorded by the mapper (read only)
`Job_Oct20/Processing/<Name>/<Name>.dcsproj`	The shared project: captures, shared settings (coordinate system, base station, antennas, Lidar mounting, vehicle) and the state of each stage. A `.bak` copy is kept beside it.
`01_PosPro/`, `02_AlignPro/`, `03_CloudPro/`	Each stage's own settings file, `work/`, `log/`, and `out/` with what it finalizes
`<Name>.qgz`	The QGIS project, when you save it from the project menu

Mapper 2.10 projects

From mapper 2.10 the mapper keeps its own projects. Use the mapper project's folder as the job folder:

PATH	HOLDS
`Deck_Test/project.json`	The mapper project: name, id, next line, operator, vehicle
`Deck_Test/lines/Deck_Test_L001_20261104_124512/`	One survey line (one Start/Stop), its files named `Deck_Test_L001_20261104_124512_0001.clraw`, `_0002` … Captures made before 2.10 that the mapper grouped into the project keep their old names here.
`Deck_Test/nav/Deck_Test_NAV_20261104_120000_0001.clraw`	A navigation recording: everything the Certus sent while the project was open (no Lidar). A new one starts at each power-on, project opening, UTC date change and every 2 GB.

If you pick the project's `lines` or `nav` folder, or a line folder, the New project dialog takes the mapper project's folder as the job folder. The survey lines are recognised from what the files record (their line number and capture), not from their names, so a line split over several files is one line and old and new captures mix freely, in recording order. Navigation recordings are never taken as survey lines.

The Processing folder can also live on another drive (choose **Elsewhere…** when you create the project). Paths are stored relative to the job folder where possible, so a job can be copied or moved as a whole.

The project menu

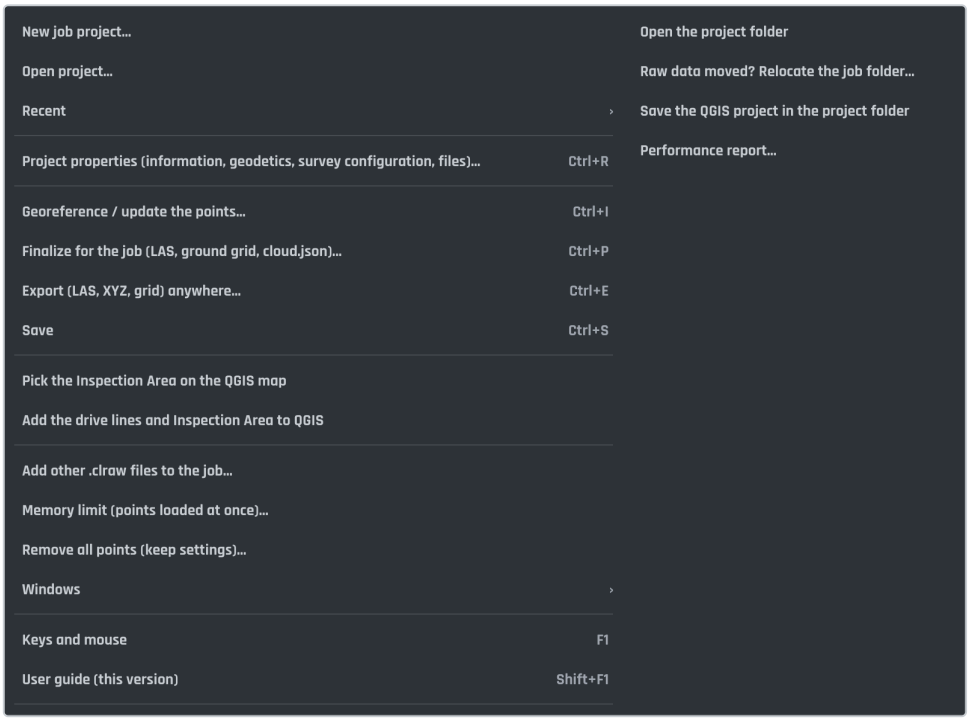


Figure 4.1 The project box menu in CloudPro.

ITEM	WHAT IT DOES
NEW JOB PROJECT…	DCS Core's New DCS project dialog (below). The new project opens in CloudPro at once.
OPEN PROJECT…	Opens a `.dcsproj`. An older stand-alone CloudPro project (`*.cloudpro.json`) can be chosen in the same dialog.
RECENT	The last ten projects opened in CloudPro.
OPEN THE PROJECT FOLDER	Opens `Processing/<Name>/` in the file manager.
RAW DATA MOVED? RELOCATE THE JOB FOLDER…	Asks for the job folder's new location and finds the captures there again.
SAVE THE QGIS PROJECT IN THE PROJECT FOLDER	Writes the current QGIS project as `<Name>.qgz` beside the `.dcsproj`.
USER GUIDE (THIS VERSION)	Opens this guide.
CloudPro's menu	Project properties, Georeference, Check line separation, Finalize for the job, Export, Save, the QGIS map tools, other captures, the memory limit, Remove all points, Windows, Keys and mouse and the user guide (Layout, menu and tools).
PERFORMANCE REPORT…	Saves the performance report to send for review (Large areas).

Creating a project

1. Choose NEW PROJECT… in the project box.
2. **Job folder**: the folder that holds the mapper's capture folders. The project name defaults to the date of the first capture with a number, for example *2026-11-03_1*; when that project exists already the next number is offered (*2026-11-03_2*), so a second processing of the same job (another base station, other settings) gets its own folder. Type any other name if you prefer.
3. **Processing folder**: inside the job folder by default; **Elsewhere…** to put it on another drive.

4. Press OK. The project is created and opened. When the job folder holds survey lines (`.clraw`; mapper 2.10 navigation recordings are not survey lines), CloudPro opens the Georeference dialog straight away (Georeferencing).

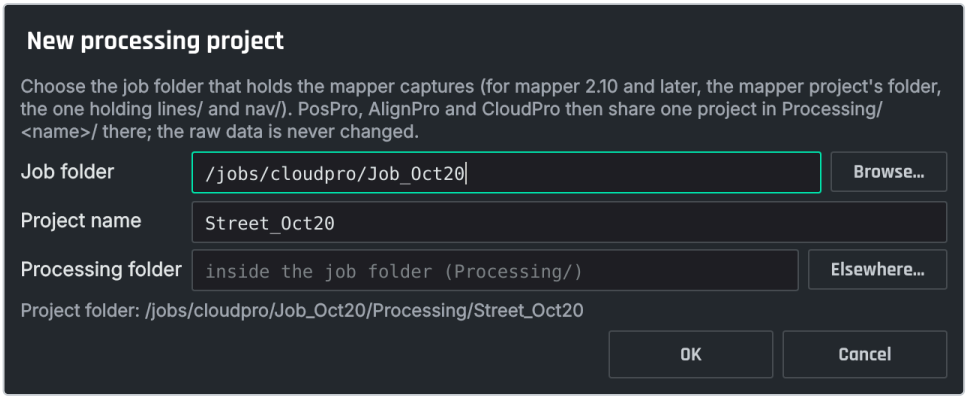


Figure 4.2 The New DCS project dialog (the same in every plugin).

A project made in any of the three plugins is the same project: open the `.dcsproj` in another plugin, or click its box in the header.

When the raw data has moved

If the captures are not where they were (another drive letter, a copied job), CloudPro says so when the project opens and asks for the job folder's new location; `RAW DATA MOVED? RELOCATE THE JOB FOLDER…` in the project menu does the same at any time. The captures are found again by their place in the job folder; nothing goes out of date because of the move.

Out-of-date results

Each stage records what it was built from. When an earlier stage finalizes again, the raw data changes, or a shared setting it depends on is edited, the stage turns **amber (out of date)** in the chain in the header, with the reason on hover. Run that stage again to bring it up to date. A run that fails or is cancelled keeps the earlier results.

5 Georeferencing

CloudPro reads only what PosPro and AlignPro finalized for the job, and records which results it used. Georeferencing builds CloudPro's point store from the captures; it is needed once per project and again whenever its inputs change.

The inputs

INPUT	USED	WHEN IT IS MISSING
Trajectory	PosPro's `01_PosPro/out/trajectory.json`: the SBET with its SMRMSG, frame, epoch and GPS week.	The Certus real-time track recorded in the capture is used, with the warning that it can drift by metres without RTK. Run PosPro, then update.
Mounting, Lidar lever arm, latency	AlignPro's `02_AlignPro/out/alignment.json`, when it uses the mapper's mounting convention.	The Lidar mounting shared in the project, else the one recorded in each capture; the lever arm recorded in the capture; no latency. Every result says *no calibration*.
Captures	The `.clraw` files in the project's capture folders (never anything under `Processing/`).	A message says no `.clraw` files were found; use **PROJECT MENU › ADD OTHER .CLRAW FILES TO THE JOB…**
Coordinate system	The job's shared horizontal CRS, else the one recorded in the capture, else EPSG:2953. Heights are ellipsoidal unless a geoid grid is given. A PosPro trajectory processed with PPP (no base station) is in ITRF2020 at the observation date; CloudPro moves it to NAD83(CSRS)v8 (the current realisation, at the observation epoch) for a NAD83(CSRS) CRS, uses it as it is for a WGS 84 CRS, and stops with a message for any other datum rather than leave the cloud metres off. **Epoch:** NAD83(CSRS) coordinates move a few millimetres a year, so they are given with an epoch; each province adopts one for its work (New Brunswick 1997.0; Nova Scotia, PEI, Newfoundland and Labrador and federal work 2010.0). Project properties › Geodetics › **COORDINATE EPOCH** chooses it: the provincial standard for the CRS (the default), 1997.0, 2010.0 or as observed. When the trajectory is in another epoch – a PPP trajectory at the observation date, or a PPK one whose base position is at the observation date – CloudPro moves it with NRCan's NAD83(CSRS) velocity grid (about 8 cm horizontally and 2 cm vertically between 2026 and 1997.0 near Fredericton). A base position entered without an epoch is used as it is, with a warning. `cloud.json` and the report give the epoch of the points.	Change it in Project Properties › Geodetics.

CloudPro also warns when PosPro's or AlignPro's results are themselves out of date, and when the AlignPro calibration was solved on an earlier PosPro trajectory.

The Georeference dialog

Open it with **PROJECT MENU › GEOREFERENCE / UPDATE THE POINTS…** (`Ctrl+I`), the **GEOREFERENCE** step in the header, or by creating a new project on a job folder with captures.

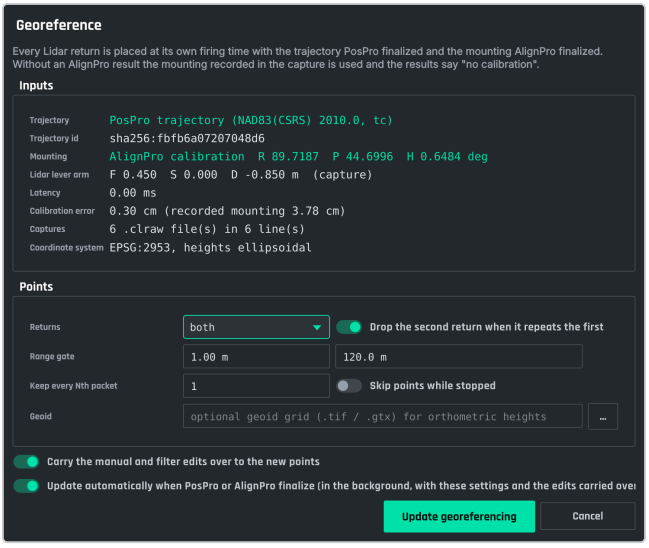


Figure 5.1 The Georeference dialog with PosPro's trajectory and AlignPro's calibration (green). On a project that already has points the button reads `UPDATE GEOREFERENCING`.

PART	WHAT IT SHOWS OR SETS
Inputs	Trajectory and its id, Mounting (roll, pitch, heading), Lidar lever arm (forward, starboard, down) and where it came from, Latency, Calibration error (AlignPro's average error, and that of the recorded mounting), the number of captures and lines, the coordinate system. Green is the chain's best input; amber is a fallback. Warnings follow in amber, each starting with **!**.
`RETURNS`	*both*, *last* or *strongest*. `DROP THE SECOND RETURN WHEN IT REPEATS THE FIRST` (on by default).
`RANGE GATE`	Returns nearer than the first value or further than the second are not imported (default 1 to 120 m).
`KEEP EVERY NTH PACKET`	1 keeps all points; N thins the import by keeping every Nth Lidar packet.
`SKIP POINTS WHILE STOPPED`	Leaves out the points recorded while the vehicle stands still (below 0.3 m/s, from the trajectory). Parked or waiting, the scanner repeats the same spot about a million times a second, which adds noise, disk space and time to every update. On for new projects; an update keeps what the lines used before until you change it.
`GEOID`	An optional geoid grid (`.tif`, `.gtx`) for orthometric heights; empty gives ellipsoidal heights.
`CARRY THE MANUAL AND FILTER EDITS OVER TO THE NEW POINTS`	Only on a project with points (on by default). Edits are matched by firing time, laser and return, so they follow the points to their new positions.

The point options of the previous georeferencing are filled in again. Press `GEOREFERENCE` (or `UPDATE GEOREFERENCING`). The work runs in the background; the status strip at the bottom of the window shows the progress and a `CANCEL` button. The new points are built beside the current ones and replace them only when everything has worked: after `CANCEL`, an error, or a trajectory that does not cover a line georeferenced before, the previous points and every edit are kept as they were. While a task runs, the actions that would change the points or the project (Undo, editing, the menu, the filters, opening another project) wait; looking around and navigating stay available.



Figure 5.2 The status strip while georeferencing: the current step on the left, the loaded point counts, the progress bar and Cancel on the right.

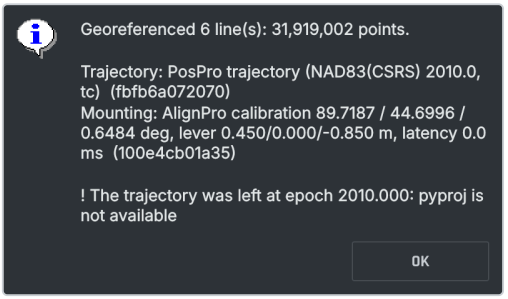


Figure 5.3 The result: lines and points, the trajectory and mounting used, and how many edited points were carried over.

> HEIGHTS The dialog's **COORDINATE SYSTEM** row follows the geoid field: *heights ellipsoidal* without a grid, *heights orthometric (geoid …)* with one. Project Properties › Project information and the report show the same afterwards.

Each run writes a log in `03_CloudPro/log/` and a line in the History tab.

Out of date: Update georeferencing

CloudPro compares the PosPro and AlignPro results its points were built with against what those stages have finalized now, every few seconds. When they differ, or when the georeferencing values in Project Properties were changed, the inputs line turns amber with the reason and the workflow strip's button reads **UPDATE GEOREFERENCING**:

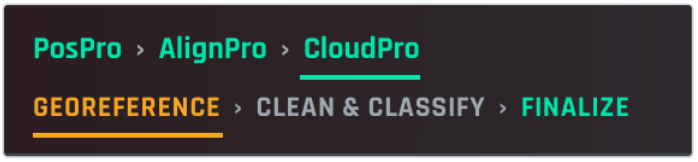


Figure 5.4 The header when the points are out of date (here after the mounting source was changed in Project Properties).

The reasons read, for example, *PosPro has finalized a new trajectory since the points were georeferenced*, *AlignPro has finalized a calibration (the points use the recorded mounting)* or *the georeferencing values in Project properties changed*. Press **UPDATE GEOREFERENCING** on the strip, keep **CARRY THE MANUAL AND FILTER EDITS OVER…** ticked, and press **UPDATE GEOREFERENCING**. Then finalize again: finalizing without updating still leaves CloudPro out of date in the chain in the header, because `cloud.json` names the results the points were really built from.

Fallbacks and your own values

When AlignPro has not finalized, or Project Properties › Survey configuration is set to **RECORDED IN THE CAPTURE (NO CALIBRATION)**, the dialog shows the recorded mounting in amber:

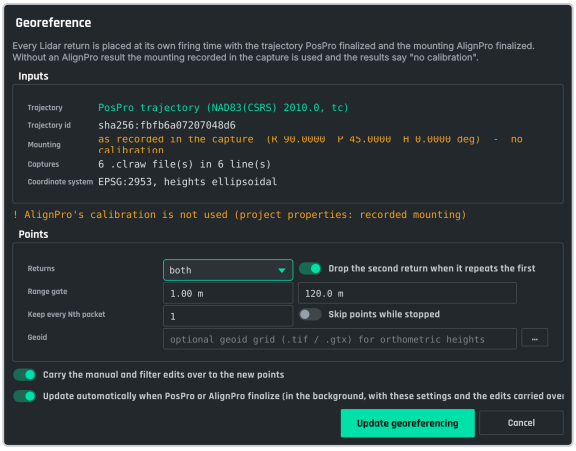


Figure 5.5 Georeferencing with the mounting recorded in the capture: *no calibration*.

With `ENTERED HERE` in the same tab, the mounting, lever arm and latency you type are used and the results say *entered values*. These values are kept in CloudPro's project; the captures are not changed. See Project Properties.

Field check: coverage before PPK

To see the extent, coverage and density of what was collected (does the area need more lines?) you do not need a post-processed trajectory, a base station or a calibration:

1. Create the project on the job folder (or the mapper project's folder) in CloudPro. Do not run PosPro or AlignPro.
2. Georeference. CloudPro uses the Certus real-time track recorded in each capture and the mounting recorded by the mapper, and says so (*no calibration*, real-time track). With RTK in the field the real-time track is usually good to a few centimetres; without it, expect metres.
3. In the Plan, set `GRID` to *Coverage* and enter the density the survey needs in the field beside it (pts/m², remembered). Set the cell size (right-click the Plan's legend) (1 m is a good start). Cells at or above the target are green, from half of it amber, below half red; empty cells show no grid. The corner gives the share of the covered area on target, the area covered and the median density. *Density* shows the same values on a colour scale.

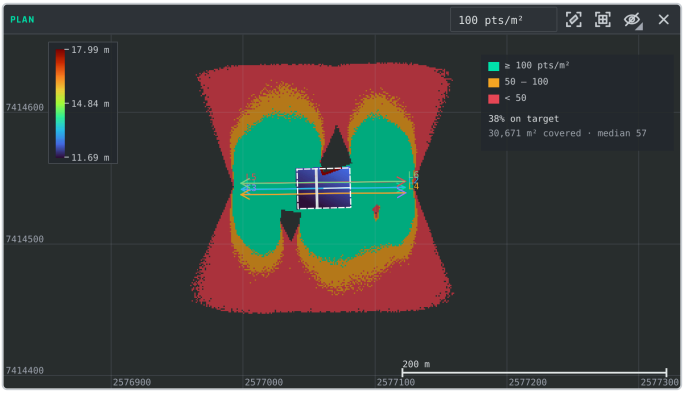


Figure 5.6 Coverage at 1 m cells with a target of 100 pts/m²: green on target, amber from 50, red below 50. The two crossing passes give the dense centre; density falls with range towards the edges. The points of the Inspection Area are drawn on top.

Density counts the points kept in the project: rejected points are left out, and a decimation or range limit set when georeferencing lowers it. From the command line, `cloudpro info` prints the same summary (`--target`, `--cell`).

When PosPro and AlignPro finalize later, the points turn out of date; the `GEOREFERENCE` step in the header re-georeferences them and keeps your edits (Out of date).

Line separation: do the lines agree?

After PosPro and AlignPro, overlapping lines should lie on top of each other. What is left (GNSS under trees or between buildings, an outage, a height step between the run out and the run back) differs from line to line and from day to day; AlignPro's calibration cannot remove it. `PROJECT MENU › CHECK LINE SEPARATION…` measures it on the points. Nothing is changed.

CloudPro cuts the project into 1 m cubes. In each cube, each line's points get a plane when they are flat and thin (ground, roofs, walls; trees and cars are left out). Where two lines both have a plane in a cube, their separation is the distance between the two planes. Flat ground and roofs give the vertical separation, walls the horizontal one. Rejected points and noise classes are left out.

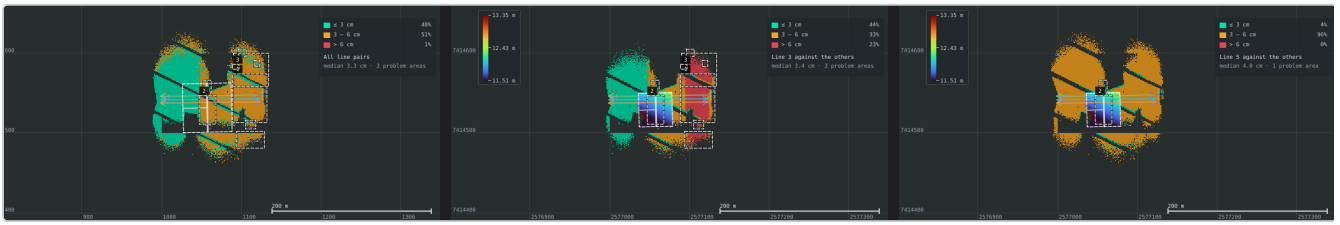


Figure 5.7 The Plan's *Line separation* grid on a job where Line 3's height drifts by up to 8 cm along the line and Line 5 is 3 cm east and 4 cm high. Left: every pair. Middle: Line 3 against the others, green at its start and red at its end (the error changes along the line). Right: Line 5, amber everywhere (one constant shift). The dashed outlines are problem areas.

When the check is done, the Plan switches to the *Line separation* grid: each 1 m cell is green when the lines there agree within the tolerance, amber within twice the tolerance, red beyond. The key gives the share of cells in each class and their median. Right-click the Plan's legend for the `TOLERANCE` (1, 2, 3, 5 or 10 cm, remembered), the results window and `CHECK LINE SEPARATION AGAIN`. Choosing *Line separation* as the grid before the check has run starts it.

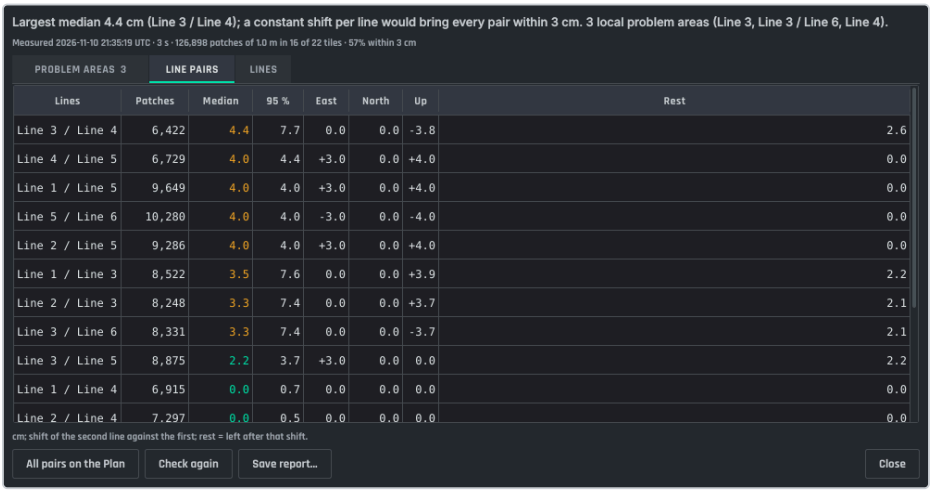


Figure 5.8 The results: one sentence on what the numbers say (with the problem areas), then the problem areas, every line pair and every line, one tab each.

COLUMN	WHAT IT IS (CM)
Patches	The number of 1 m cubes where both lines have a plane.
Median, 95 %	Of the separation over those patches (coloured like the grid).
East, North, Up	The one shift of the second line against the first that best explains the separations. East and North need walls facing in different directions; without them they show –.
Rest	The median separation left after that shift. Small: one shift per line would fix it. Large: the error changes along the line.

STEADY OFFSETS: CALIBRATION OR TRAJECTORY

Two lines often sit a steady few centimetres apart over their whole overlap. CloudPro solves that steady offset per pair in the vehicle's own frame, from each line's direction of travel: along the road, sideways, up, and a roll slope (a height difference that grows with the distance from the drive lines). What it means depends on how the lines were driven:

- **Both ways**: a calibration error moves each pass's points the same way relative to its own travel, so it shows as an offset that reverses with the driving direction (each pass carries half of it). Along the road: timing latency, an along-track lever arm or pitch; sideways: a cross-track lever arm or heading; a growing height difference: roll. These point at **AlignPro**.
- **The same way**: a calibration error cancels, so a steady offset means the two trajectories disagree (**PosPro**).

Along the road and sideways are measured on walls and other steep surfaces only: on gently sloping ground a height difference between the passes would otherwise look like a shift along the road. The along-road offset is also measured in four parts of the line; when it changes from part to part it is a trajectory error, whatever its size, because a calibration error is the same everywhere.

The `STEADY OFFSETS` tab lists each pair with what it points at; the sentences under it say why for the chosen pair, and the first sentence of the window names the main one. Each check writes them to `out/calibration_check.json`, where **AlignPro** shows them on its Pairs page and can calibrate on that pair's lines and overlap. Right-click the Plan's legend › `STEADY OFFSETS REMOVED` to see what is left once each pair's steady offset is taken off.

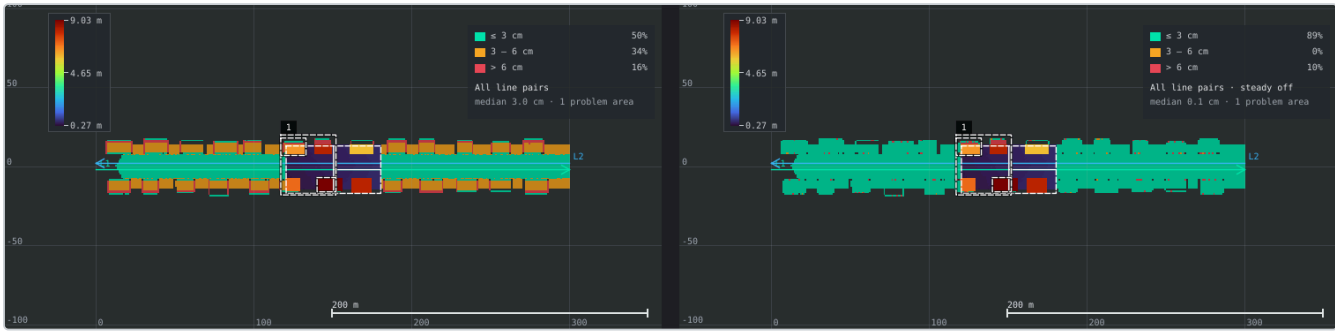


Figure 5.9 A street driven both ways with each pass 4 cm ahead of the vehicle, a roll error, and a few seconds 12 cm low on the way back. Left: as measured, the steady offset makes the street amber. Right: steady offsets removed, only the real problem (1) is left.

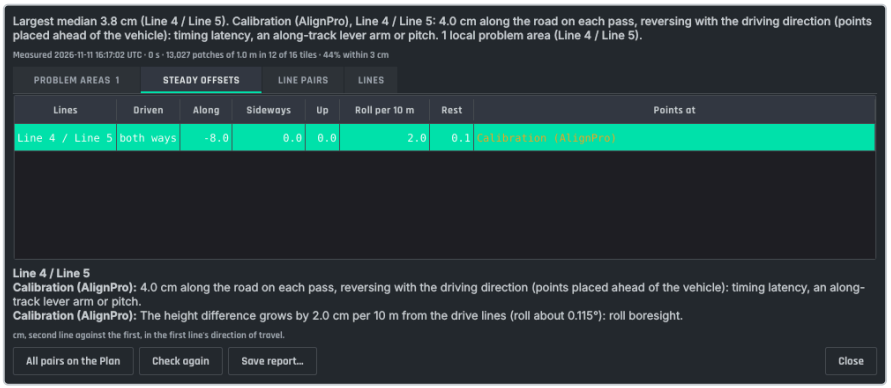


Figure 5.10 The STEADY OFFSETS tab for that street: −8.0 cm along (4 cm each pass) and 2 cm per 10 m of roll, pointing at the calibration.

PROBLEM AREAS

A trajectory error that lasts a few seconds (a wrong fix under a bridge, a GNSS outage) hardly moves a pair's median, but shows on the Plan as a red patch on one line. CloudPro finds these on what is left after each pair's steady offset (a steady offset is one finding, not hundreds of areas): clusters of cells beyond twice the tolerance, at least 10 m² and less than half of the pair's overlap. When the patch shows against several lines, the line they share is the one at fault; pieces of the same line (or the same pair) up to 3 s apart are one area. Each area has a number on the Plan (dashed outline) and a row under `PROBLEM AREAS`: the line, where, its size, the median and largest separation, and the time its points were taken (UTC, and GPS seconds of the week as in a trajectory). When the line at fault cannot be told (only two lines there), both lines and their

times are given. Click a row: the Plan shows that line against the others and the Inspection Area goes onto the area, along the drive line, for a look in the Slice and the 3D view.

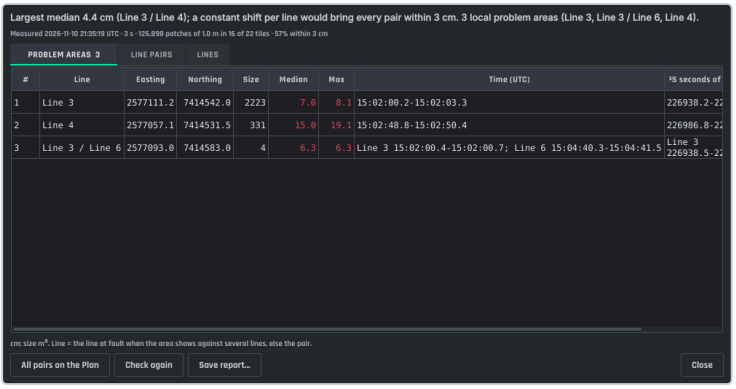


Figure 5.11 Problem areas on the test job: Line 4 15 cm off for 2 s (2), the end of Line 3's drift (1), and a small patch where only Lines 3 and 6 overlap (3).

The first sentence mentions the problem areas even when the medians look fine. Each check also writes them to `out/problem_areas.json`, where **PosPro** reads them: they appear in its Trajectory Review and Trajectory editor as *Point cloud* windows, to bridge with the IMU, mark poor, interpolate or accept. After PosPro finalizes the corrected trajectory and CloudPro updates the points, check again: the area should be gone.

Click a pair or a line to show only it on the Plan; **ALL PAIRS ON THE PLAN** shows every pair again. **SAVE REPORT…** writes the same as a text file; each check is also kept in the History and in `log/`. The result is kept with the project and marked *out of date* when the points are georeferenced again. The check reads every point once: about as long as reading the project from the disk.

Other captures, removing points

- **PROJECT MENU › ADD OTHER .CLRAW FILES TO THE JOB…** picks `.clraw` files, asks *Add N file(s) to the job's captures? The captures are shared by PosPro, AlignPro and CloudPro.*, adds them to the shared project and opens the Georeference dialog.
- **PROJECT MENU › REMOVE ALL POINTS (KEEP SETTINGS)…** deletes the point store and the edits after a confirmation. The captures are not touched; Georeference rebuilds the points. Filter groups, the coordinate system and your entered values are kept.

Older stand-alone projects

A project from CloudPro 1.0 (`<name>.cloudpro.json`) still opens, without the chain. For it, Georeference and Add other .clraw files open the import dialog, where you choose the inputs yourself; finalizing is not available (use Export).

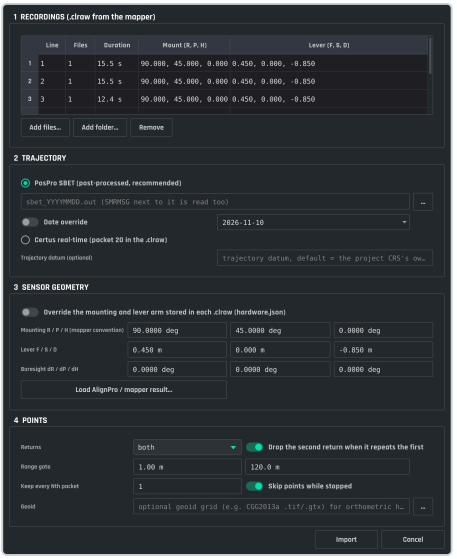


Figure 5.12 The import dialog of an older stand-alone project, with two captures added.

GROUP	SETTINGS
1 Recordings	**ADD FILES…** , **ADD FOLDER…** (all `.clraw` below it), **REMOVE** . One row per recording: line, files, duration, recorded mounting and lever arm.
2 Trajectory	**POSPRO SBET (POST-PROCESSED, RECOMMENDED)** with the file (an SBET beside the captures is filled in; the SMRMSG next to it is read too) and **DATE OVERRIDE** for SBET names without a date; or **CERTUS REAL-TIME (PACKET 20 IN THE .CLRAW)** . Optional trajectory datum.
3 Sensor geometry	**OVERRIDE THE MOUNTING AND LEVER ARM STORED IN EACH .CLRAW** , mounting R / P / H, lever F / S / D, boresight dR / dP / dH; **LOAD ALIGNPRO / MAPPER RESULT…** reads them from a JSON file.
4 Points	Returns, duplicate second returns, range gate, every Nth packet, geoid, as in the Georeference dialog.

CloudPro's folder

PATH IN `03_CLOUDPRO/`	HOLDS
`cloudpro.json`	CloudPro's settings: lines, inputs used, view, history, filter runs, entered values.
`filters/groups.json`	The filter groups of this project.
`work/store/`	The point store in 50 m tiles and the project grids. A cache of the raw data: it can be deleted, Georeference rebuilds it (the edits are lost then).
`work/lines/`	The sensor track of each line.
`out/`	What Finalize delivers (Finalizing).
`log/`	One log per run (georeference, filter, tune, finalize).

While a job's cloud is open in CloudPro it is locked for editing: a second QGIS that opens the same project is told *This job's point cloud is open elsewhere* and does not open it. Georeferencing, project-wide filter runs and finalizing also hold the stage lock, which shows CloudPro as **running** in the chain in the header.

6 Layout, menu and tools

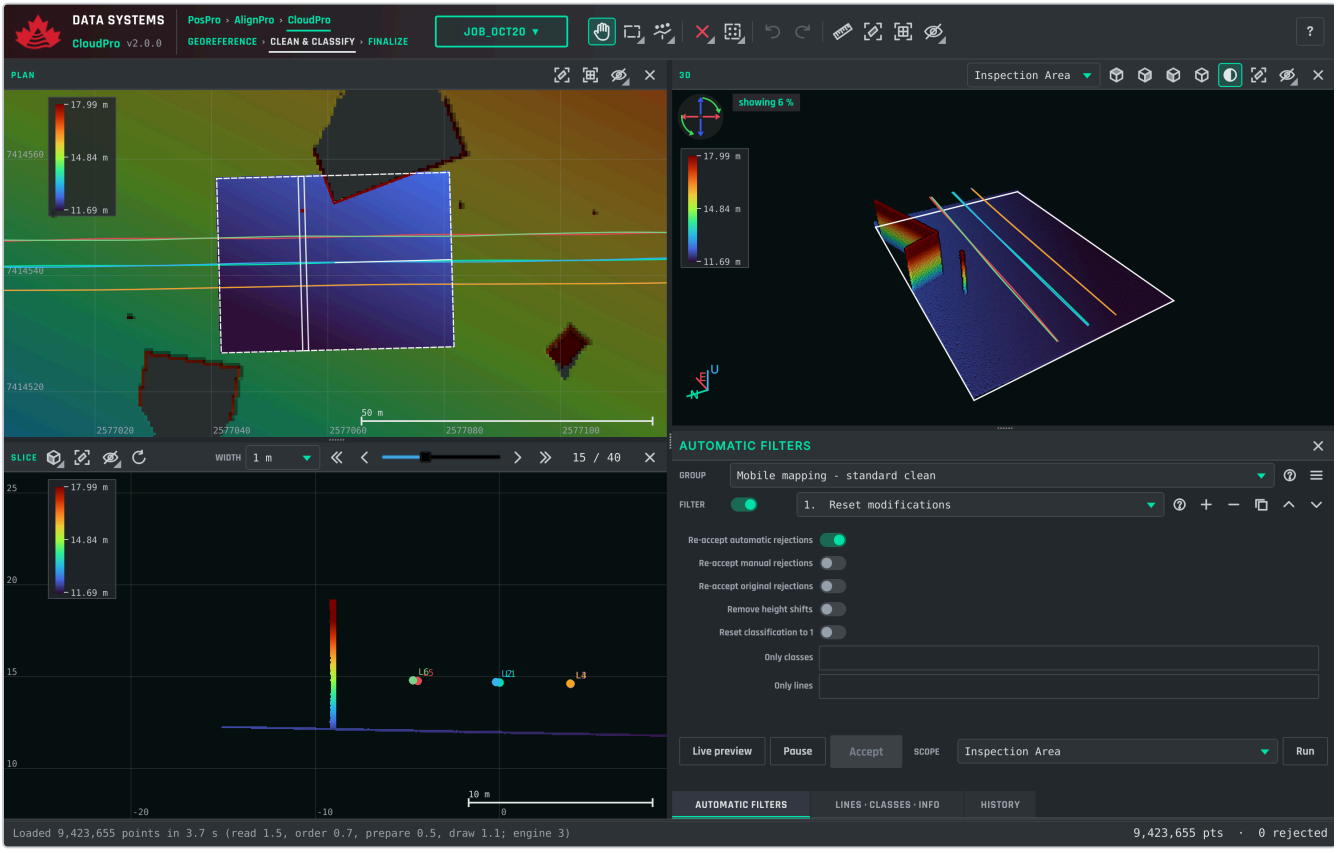


Figure 6.1 CloudPro on a job: the header with its steps, project box and edit tools, the Plan, Slice and 3D views, the tabbed Automatic filters, Lines · Classes · Info and History, and the status strip.

PART	WHAT IT HOLDS
Header	The chain and CloudPro's steps, the project box (with CloudPro's menu), the edit tools and the user guide button (chapter 3).
Plan (top left)	The whole project as a grid, the drive lines, the Inspection Area and its points from above.
Slice (bottom left)	A slab of the Inspection Area from the side: a cross-section or a profile.
3D (top right)	The Inspection Area, or only the slice, in 3D.
Tabs (bottom right)	`AUTOMATIC FILTERS`, `LINES · CLASSES · INFO`, `HISTORY`, tabbed together.
Status strip	Messages on the left; the cursor position (E and N in the Plan; position and Z in the Slice); the points loaded, rejected and selected; the progress bar and `CANCEL` while a task runs.

All three views show only the points inside the **Inspection Area**, the box you work in; the Plan's background grid is the only thing drawn from the whole project.

Views and panels

The views and panels stay inside CloudPro's window: they do not float or move. Drag the line between two of them to resize them. The × on a view's or panel's title strip hides it; `PROJECT MENU › WINDOWS` shows it again, and `WINDOWS › RESET THE WINDOW LAYOUT` brings back the default layout. The sizes are remembered on this computer. A view

that is hidden, or behind another tab, is not drawn while you work, so hiding the 3D view when you do not need it makes every edit and every move faster; it catches up when it is shown again.

CloudPro's menu

CloudPro's actions are in the project box's menu (click the project name in the header), under the project items.

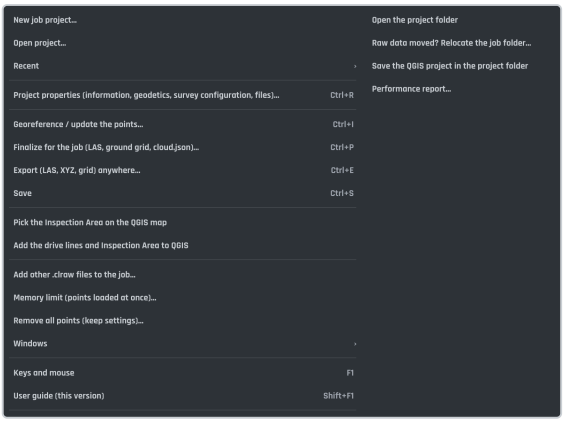


Figure 6.2 The project box's menu.

ITEM	KEY	DOES
Project properties (information, geodetics, survey configuration, files)...	`Ctrl+R`	Project Properties
Georeference / update the points...	`Ctrl+I`	Georeferencing
Check line separation...		Line separation
Finalize for the job (LAS, ground grid, cloud.json)...	`Ctrl+P`	Finalizing
Export (LAS, XYZ, grid) anywhere...	`Ctrl+E`	Export
Save	`Ctrl+S`	Saves the project, the view and the filter groups. Edits are written to disk as you make them, and CloudPro saves quietly every 2 minutes.
Pick the Inspection Area on the QGIS map		QGIS layers
Add the drive lines and Inspection Area to QGIS		
Add other .clraw files to the job...		Georeferencing
Memory limit (points loaded at once)...		Large areas
Remove all points (keep settings)...		Georeferencing
Windows		Show or hide each view and panel; **RESET THE WINDOW LAYOUT**.
Keys and mouse	`F1`	A summary of the keys and mouse actions (Reference).
User guide (this version)	`Shift+F1`	This guide.

The edit tools

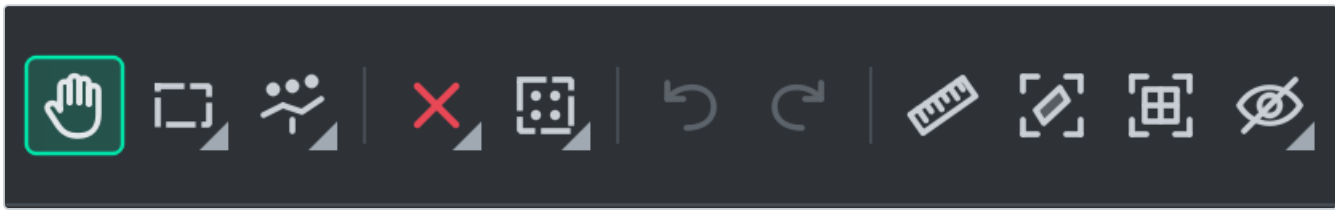


Figure 6.3 The edit tools in the header, right of the project box: tools, action, inside / outside, undo / redo, measure, zoom and rejected points.

BUTTON	OPTIONS (KEY)
Hand	Pan (`Space`): drag to move the view; in the Plan, move, resize and rotate the Inspection Area.
Selection shape	Rectangle (`G`), Polygon (`P`), Lasso (`L`), Brush (`B`).
Line	Above a line (`U`), Below a line (`J`).
Action	Reject (`X`), Accept (`V`), Select (`S`), Classify (`K` , with the class list next to it), Filter (`F` : runs the current filter group on the shape). The shape you draw (rectangle, polygon, lasso, brush, above / below) shows the action: **red** for Reject, **green** for Accept, **grey** for Select (teal for Classify and Filter).
Inside / outside	Edit the points inside the shape, or the visible points outside it.
Class list	Shown with Classify: the class it sets (1–7, 9, 11, 14, 18, 64–71).
Undo, Redo	`Ctrl+Z` , `Ctrl+Y` . The Undo tooltip names the step it would undo.
Ruler	Measure (`M`): distance, bearing, height difference, slope.
Zoom to the Inspection Area	All views (`Ctrl+Home`).
Zoom to the extents	The Plan shows the whole project.
Eye	Rejected points hidden, or shown in magenta, in every view at once (`H`). Each view also has its own eye in its title strip.

Edits change every line. To edit one line only, right-click it in **LINES · CLASSES · INFO** and choose **EDIT ONLY THIS LINE** (**EDIT ALL LINES** goes back). Point information: press `I` and click a point. The point size, the Plan's cell size and the brush size are in the right-click menu of each view's legend (Colours).

Buttons with a small triangle in the corner hold several options. The first click on an inactive one makes it active; each further click switches to its next option. Click the corner triangle, or right-click the button, for the list:



Figure 6.5 The option lists of the selection shape, line, action and inside / outside buttons.

Choosing an action while the hand, ruler or i is active switches to the last selection shape, ready to draw. Hover any button for its name and key.

7 Views and navigation

Linked cursor

Move the mouse over a point in any view and the other two views show the same point: a white **cross** on it in the Plan and the 3D view, and a dashed **vertical line** at its place across the slice in the Slice view (whether the point is in the current slice or not). Over empty ground in the Plan, which has no height there, the 3D view shows a vertical line too. In the Plan the point is the one drawn under the mouse (the top one); in the Slice the nearest one; in the 3D view the nearest visible one, within 10 pixels. The marks go when the mouse leaves the view.

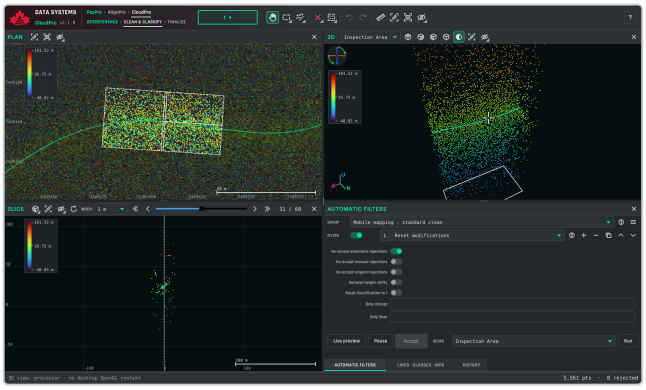


Figure 7.1 The mouse on a point in the Plan: a cross marks it in the 3D view, a dashed vertical line in the Slice.

Plan

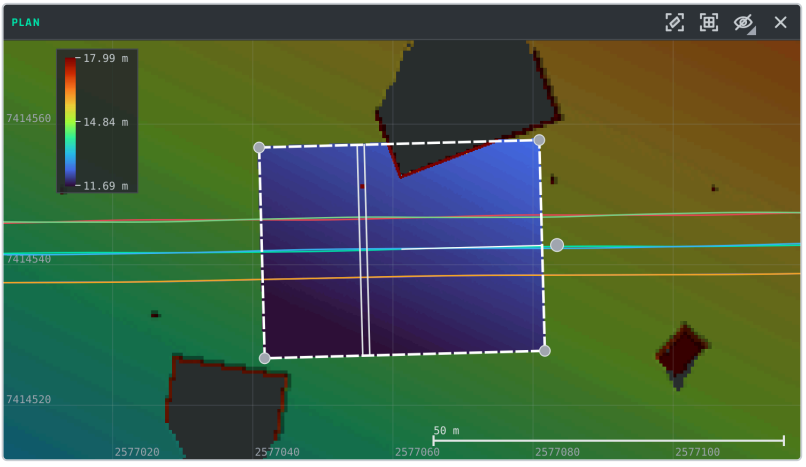


Figure 7.2 The Plan: the project grid (mean height, hill shaded), the two drive lines L1 and L2, the Inspection Area with its handles under the mouse, the two lines of the slice, and the legend of the points (left).

The Plan's title strip has `ZOOM TO THE INSPECTION AREA`, `ZOOM TO THE EXTENTS` and the rejected points eye. Right-click its legend for `GRID` (the background of the whole project: *Mean height*, *Max height*, *Min height*, *Span*, *Hit count*, *Density* (pts/m²), *Coverage* against a target density (see Field check), *Line separation* (Line separation), *Reject count*, *Intensity* or *None*), the cell size and the brush size, besides the colours every view has (Colours). The grid has no colour scale of its own: its colours only show shape and change (one legend per view, for the points); *Coverage* and *Line separation* show their key. The grid is dimmed so the points of the Inspection Area stand out on top; heights are hill shaded. Edits update the grid within a few seconds.

Each drive line is drawn in its line colour, labelled at its start (L1, L2…) with an arrow at its end. The slice is shown as two thin lines across or along the Inspection Area.

THE INSPECTION AREA

With the Pan tool:

- Drag the dashed **edge** to move the area, a **corner** to resize it (in whole metres), the **round handle** beyond its front to rotate it. The handles appear when the mouse is over the area; while you drag, its centre, size and bearing are shown. The points are loaded when you release.
- `Shift` + drag anywhere draws a new area along the drag, keeping the width.
- Drag elsewhere to pan the view; the wheel zooms at the cursor.

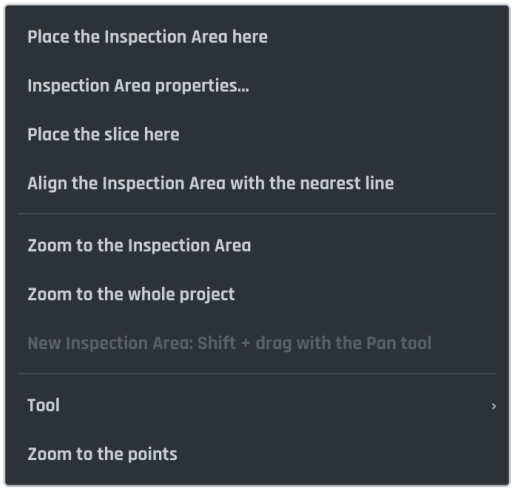


Figure 7.3 Right-click in the Plan.

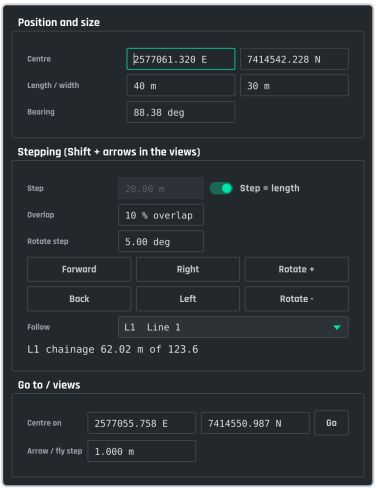


Figure 7.4 The Inspection Area by numbers, here following line L1.

Right-click in the Plan for **PLACE THE INSPECTION AREA HERE**, **INSPECTION AREA PROPERTIES…**, **PLACE THE SLICE HERE**, **ALIGN THE INSPECTION AREA WITH THE NEAREST LINE** (centred on the nearest drive line and turned along it), **ZOOM TO THE INSPECTION AREA**, **ZOOM TO THE WHOLE PROJECT**, the **TOOL** list and **ZOOM TO THE POINTS**. With a selection, a **SELECTION** submenu comes first (Manual editing).

INSPECTION AREA PROPERTIES

This window stays open beside the views. Every change applies at once.

GROUP	SETTINGS
Position and size	Centre E and N, length and width (whole metres), bearing of the long axis (degrees clockwise from grid north).
Stepping (`Shift` + arrows in the views)	**STEP** in metres, or **STEP = LENGTH**: forward and back by the length, left and right by the width, less the **OVERLAP** %. **ROTATE STEP** in degrees. Buttons **FORWARD**, **BACK**, **RIGHT**, **LEFT**, **ROTATE +**, **ROTATE -**.
FOLLOW	*Free (straight steps)* or a drive line: the area moves onto the line, steps along its chainage and turns with it. The chainage is shown below (*L1 chainage 51.62 m of 123.6*).
Go to / views	**CENTRE ON** E and N, then **GO**. **ARROW / FLY STEP**: the distance the arrow keys pan the 2-D views and W / A / D / Q / E move the 3D view.

In the Lines list, right-click a line and choose **INSPECTION AREA FOLLOWS THIS LINE** for the same.

Slice

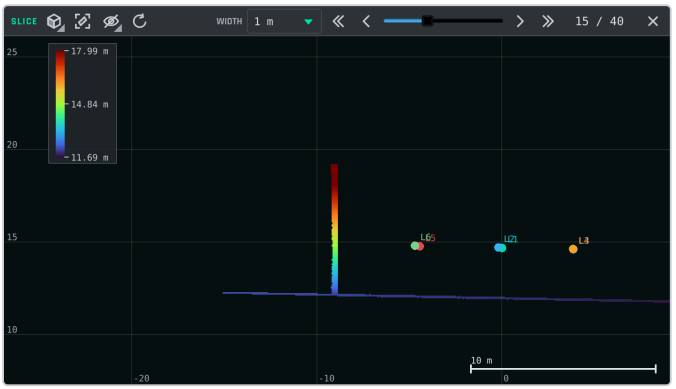


Figure 7.5 A 1 m cross-section through a pole. The dot marked L1 / L2 is where the drive lines cross the slice, at sensor height.

The Slice view shows a slab of the Inspection Area from the side: a **cross-section** across the area, looking along it, or a **profile** along the area, seen from the side. The slices are **whole slices of the chosen width inside the Inspection Area**, centred in it: they never reach past its edges. The title strip of the view:

CONTROL	DOES
Direction button	Cross-section or profile (click to switch).
Zoom, eye	Zoom to the Inspection Area; rejected points shown or hidden in this view.
Rotate	Rotate the slice 90 degrees (cross-section ↔ profile).
WIDTH	0.5, 1, 2, 4, 8 or 16 m. The number of slices is the number of whole slices of that width that fit in the area's length (cross-sections) or width (profiles); what is left over is shared equally at both ends. A width larger than the area gives one slice as wide as the area. Changing the width keeps the slice nearest the same place.
« ‹ slider › »	First slice, previous slice, the slice in the area, next slice, last slice.
12 / 36	The slice shown and the number of slices in the area at this width.

Step with `PgUp` / `PgDn`, `]` / `[` or `.` / `,`; `Ctrl+PgUp` / `Ctrl+PgDn` go to the last / first slice. Stepping stops at the area's first and last slice: move the Inspection Area on to see further. Right-click in the Slice for **ROTATE THE SLICE 90 DEGREES**, **PREVIOUS SLICE** and **NEXT SLICE**; **PLACE THE SLICE HERE** in the Plan goes to the slice under the click.

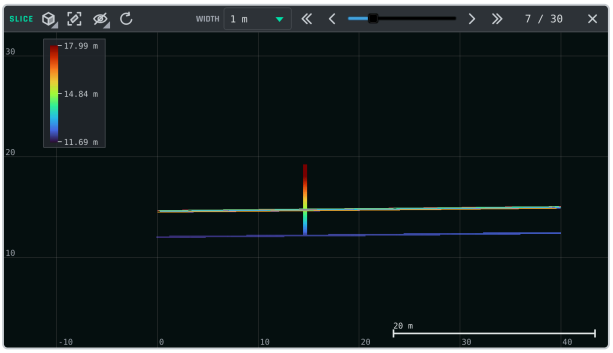


Figure 7.6 A profile along the area through two poles. In a profile the drive lines are drawn at sensor height.

3D

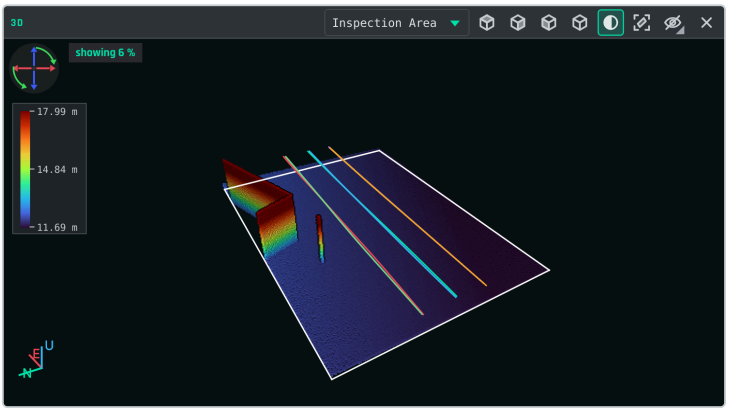


Figure 7.7 The 3D view, oblique: the navigation arrows top left, the legend, the Inspection Area outline, the drive lines and the axes (E, N, U) bottom left.

The header has what to show (*Inspection Area* or *Slice*), four view presets (top `1`, look across to see the side `2`, look along the area `3`, oblique `4`), eye-dome lighting (depth shading, on by default), zoom to the Inspection Area and the rejected points eye; the colours are in the right-click menu of its legend.

Navigation arrows. Click an arrow head to turn the cloud that way (its near side moves in the arrow's direction): **blue** tips it up or down (into the page), **red** turns it left or right, **green** turns the picture clockwise or anticlockwise on the screen. Each click turns 22.5° (16 clicks round); hold the button to keep turning smoothly. The presets undo any on-screen turn.

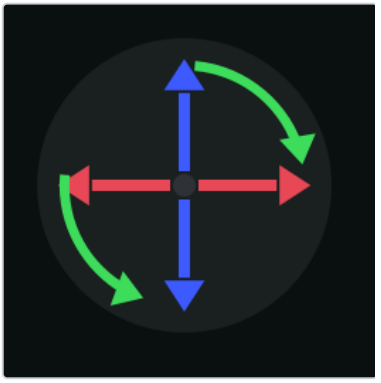


Figure 7.8 The navigation arrows.

Centre of rotation. The 3D view turns around the centre of the Inspection Area, at the middle height of its points, and the centre follows the area when you move or step it. Panning or double-clicking a point moves the centre for a closer look; the **ZOOM TO THE INSPECTION AREA** button in the 3D header (corner brackets around a rectangle), `0` or `Home` bring it back and show the whole area.

Mouse. Left drag turns the cloud the way you move, as if holding it (drag right and its near side moves right); `Shift` + left drag or middle drag pans; the wheel zooms; `Ctrl` + wheel scales Z (shown as *Z x2.0*); double-click a point to make it the centre (its information appears in the Info tab); right drag draws a lasso, or a rectangle when the Rectangle tool is active, and applies the current action.

Keys (click the view first): `W` / `A` / `D` fly forward, left and right and `Q` / `E` down and up, by the arrow / fly step; `O` switches orthographic projection; `0` or `Home` zooms to the Inspection Area; `+` / `−` zoom; `Esc` cancels a selection.

> **FLYING** `W` `S` `A` `D` `Q` `E` fly while the 3D view has the keyboard: click in it first. Elsewhere `S` is the key for the Select action.

Performance report. CloudPro writes the time of every step to `03_CloudPro/log/performance.log` in the job: each Inspection Area load (with where the time went: read, order, prepare, draw), each filter run and preview (with every filter's time), each job (georeferencing, also the automatic one, tiled filters, finalizing, exporting), and the 3D view (graphics card or processor, and its average frame times). Each time a job is opened the log starts with this computer: processor and cores, memory, QGIS, DCS Core and engine versions. To send it for review, choose **PERFORMANCE REPORT...** on the project menu (the job name in the header): it saves one text file, on the Desktop by default, with the computer, the job, the log, the recent History and the latest run logs. Nothing is sent anywhere by CloudPro itself.

Graphics card. When the computer has a graphics card with OpenGL 2.1 or later, the 3D view copies the loaded points to the card once and the card draws every point in each frame, while you move and when you stop; the eye-dome shading of a still frame is computed by the CloudPro engine on every core. When the window opens the job, the status bar says *3D view: graphics card* with the card's name, or *3D view: processor* and why (no OpenGL, a software OpenGL driver, or the

card could not hold the points). On the processor the view works on any computer: while you move it draws fewer points to stay fluid, and a still frame that would take more than about 0.3 s draws a part only and says *showing N %* beside the title (Large areas). The picture is the same either way.

Colours, legends and sizes

Each view has one legend (top left): the colour ramp with its values, or the classes, lines or files in the area. **Right-click the legend** for the view's display settings:

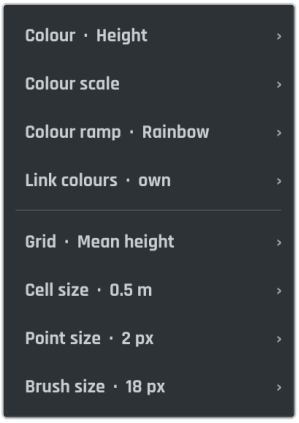


Figure 7.9 Right-click on the Plan's legend.

ITEM	SETS
`COLOUR`	What colours the points in this view (below). Greyed out while the view is linked.
`COLOUR SCALE`	For Height, Intensity, Range and Time: automatic over the Inspection Area, automatic over the whole project, or a fixed minimum and maximum (below).
`COLOUR RAMP`	Rainbow, Viridis, Terrain, Mint or Grey, for height, intensity and range in this view (the Plan's also colours its grid). A linked view uses the ramp of the view it follows.
`LINK COLOURS`	Own colours and scale, or the same as another view (below).
`GRID` (Plan)	The background grid of the whole project.
`CELL SIZE` (Plan)	The grid's cell: 0.5, 1, 2, 2.5, 5, 10, 12.5 or 25 m (the stored 0.5 m grid summed into larger cells).
`POINT SIZE`	1 to 6 pixels, in this view.
`BRUSH SIZE` (Plan)	The brush radius in pixels, for the Plan and the Slice.

COLOUR	POINTS COLOURED BY
Height	Height, with the ramp chosen in `COLOUR RAMP`.
Intensity, Intensity (range normalised)	Intensity (grey when the ramp is Rainbow), or intensity × (range / 10 m)².
Classification	The class colour (classes table).
Line, File	The line's colour, or one colour per `.clraw` file.
Return	Green last or single return, amber strongest.
Channel	The laser (1–32).
Time, Range	Firing time (UTC), distance from the Lidar.

On top of any scheme: rejected points are magenta (where a view shows them); in a filter preview, orange points will be rejected and green ones re-accepted; selected points are white.

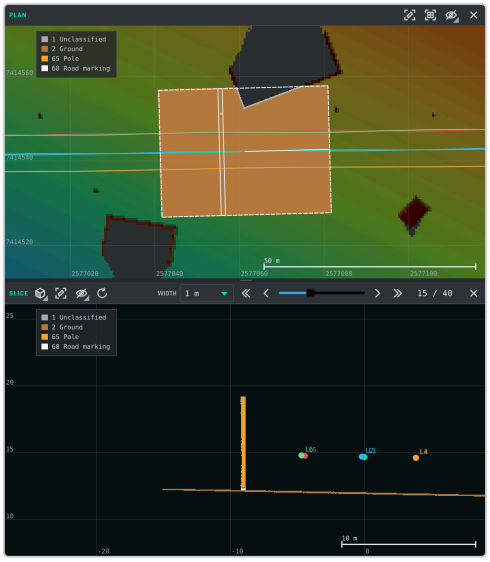


Figure 7.10 Plan coloured by classification, with the Slice linked to it (`LINK COLOURS › SAME AS THE PLAN VIEW`). Each legend lists the classes in the Inspection Area.

- **Legend**: a ramp with values (absolute heights in metres, intensity, UTC time, range or laser), or the classes, lines or files present, followed by rows for preview and selected points (rejected points are always magenta, so they have no row).
- **Colour scale**: right-click a view's legend (Height, Intensity, Range or Time colours) and choose `AUTOMATIC: INSPECTION AREA` (the default: 2–98 % of the area's points, the same in every view), `AUTOMATIC: WHOLE PROJECT` (an even sample of up to 2 million accepted points from every tile, so colours mean the same wherever the area is) or `FIXED MINIMUM / MAXIMUM…` (heights as elevations, times in UTC; points outside get the end colours). The menu shows which is chosen; each view keeps its own choice per colour scheme, saved with the project.
- **Link colours**: own colours, or the same as the Plan, Slice or 3D view: the view follows that view's scheme and colour range, whatever range it has itself. A view cannot follow one that follows it.

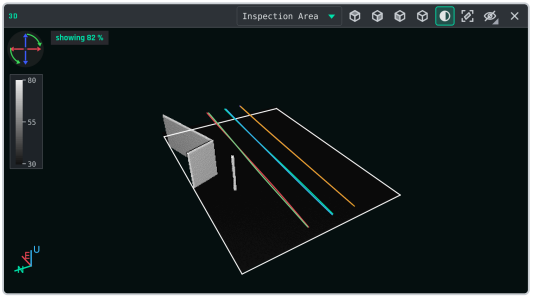


Figure 7.11 The 3D view coloured by intensity.

What is shown

- **Lines**: untick a line in `LINES · CLASSES · INFO` to hide its points in every view.
- **Classes**: untick a class there to hide it.
- **Files**: Project Properties › Project files hides a file's points (`POINTS`) or its drive line (`TRACK`).
- **Rejected points**: hidden by default. The eye in a view's title strip shows them in magenta in that view only; the eye with the edit tools or `H` sets every view. Choosing the *accept* action shows them everywhere, so they can be selected.

Hidden points are not edited, by hand or by a filter run from a shape.

2-D view controls

Zoom buttons. Every view has **ZOOM TO THE INSPECTION AREA** in its header (corner brackets around a rectangle): the Plan centres on the area, the Slice shows the area's full width (cross-section) or length (profile), the 3D view centres its rotation on the area. The Plan also has **ZOOM TO THE EXTENTS** (corner brackets around a grid): the whole project. The zoom button with the edit tools (`Ctrl+Home`) zooms all three to the area at once.

In the Plan and the Slice: the wheel zooms at the cursor; middle drag pans with any tool; double-click centres there; `Home` zooms to the points; `+` / `−` zoom; the arrow keys pan by the arrow / fly step; `Shift` + arrow keys step the Inspection Area (forward, back, left, right) and `Shift+PgUp` / `Shift+PgDn` step it forward and back. Grid labels and the scale bar are in project coordinates.

Large areas

- **Memory check.** Before an Inspection Area is loaded, its number of points is estimated from the project grids. Above the limit (15 million points by default; **PROJECT MENU › MEMORY LIMIT (POINTS LOADED AT ONCE)…**, 1 to 500 million) CloudPro asks first. If you decline, the status strip says *Not loaded: make the Inspection Area smaller*.
- **Point budgets.** While you pan, zoom or orbit, each view draws only as many points as keep it near 30 frames per second, measured from its own drawing times. The points are loaded in random order, so the part drawn is an even sample. Selecting and editing always use every point.
- **Tiles.** The point store is made of 50 m tiles; only the tiles that touch the Inspection Area are read.
- **The CloudPro engine.** Georeferencing, the tile grids and the heaviest filters (Moving objects, the cluster-based feature filters, Vertical plane) run in a small compiled library that comes with CloudPro (`bin/windows/cpengine.dll`, `bin/linux/libcpengine.so`), on every core but one, so QGIS stays responsive. It gives exactly the same points and filter results as CloudPro's Python code; the History log says which one did the work (*engine 1* or *Python*). The engine also draws the Plan, Slice and 3D views (the 3D view on the graphics card when there is one) and tests which points are inside the Inspection Area. CloudPro loads a copy of it kept in your user folder (`%LOCALAPPDATA%\CloudPro\engine` on Windows), so the plugin folder is never locked and the plugin manager can always upgrade or remove CloudPro. If the library cannot be loaded, CloudPro carries on with the Python code, only slower.

8 Manual editing

1. Choose what the shape does with the action button: Reject (`X`), Accept (`V`), Select (`S`), Classify (`K` , then the class in the list next to it) or Filter (`F`).
2. Choose *inside* or *outside*, and in the line list *All lines* or one line.
3. Choose a shape and draw it in the Plan or the Slice, or right-drag in 3D.

SHAPE	HOW TO DRAW IT
Rectangle	Drag a box.
Polygon	Click the corners; right-click, `Enter` or double-click to finish. `Backspace` removes the last corner.
Lasso	Draw freehand while holding the button.
Brush	Paint over the points; the circle shows the brush radius (**BRUSH SIZE** in the right-click menu of the Plan's legend).
Above / below a line	In the Slice, click the points of a line; right-click, `Enter` or double-click to finish. Everything above (or below) the line, between its two ends, is edited. The side that will be edited is shaded while you draw.

`Esc` cancels a shape. The points edited are the visible points inside the shape, or with *outside* the visible points outside it.

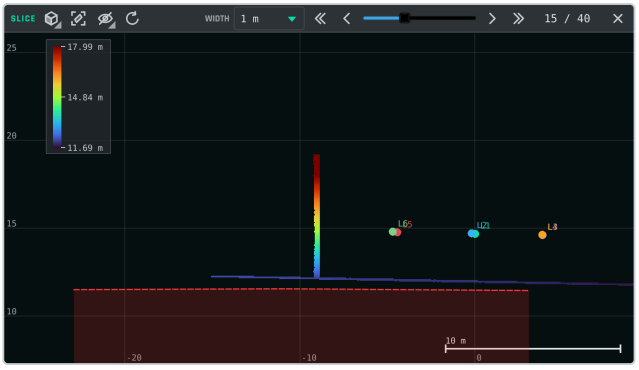


Figure 8.1 Drawing a line with *Below a line* in the Slice: the shaded side below the dashed line will be edited when the line is finished.

Selecting

With Select, the points in the shape turn white and the status strip gives their number; hold `Shift` to add to the selection. Then right-click a view:

REJECT THE SELECTION , **ACCEPT THE SELECTION** , **CLASSIFY THE SELECTION AS** (a class), **RUN THE FILTER GROUP ON THE SELECTION** , **CLEAR THE SELECTION** . `Delete` rejects the selection; `Ctrl+D` clears it.

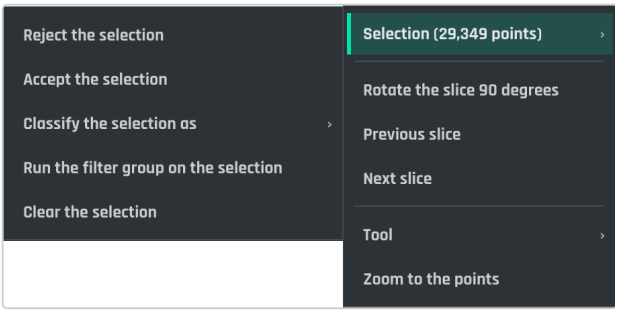


Figure 8.2 Right-click in the Slice with a selection: the Selection submenu, then the Slice's own items, the Tool list and Zoom to the points.

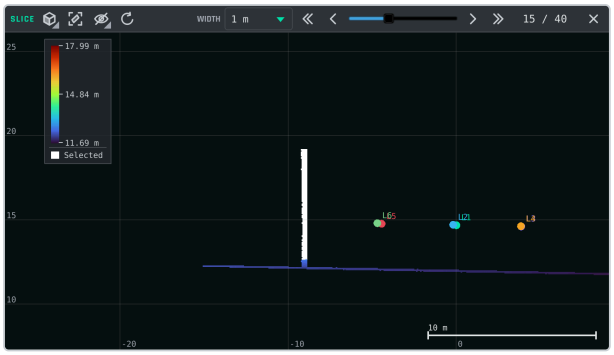


Figure 8.3 A pole selected in the Slice (white).

Filter on a shape

With the Filter action, the current filter group of the Automatic filters tab runs on the points in the shape only (all loaded points still count as neighbours), and its result is applied at once. This is a quick way to clean one spot with, for example, an outlier filter.

Measure and point information

With the ruler (`M`), click two points. In the Plan it gives the distance and the bearing; in the Slice the horizontal and vertical differences, the slope in % and the distance. The result also appears in the status strip; `Esc` clears it.

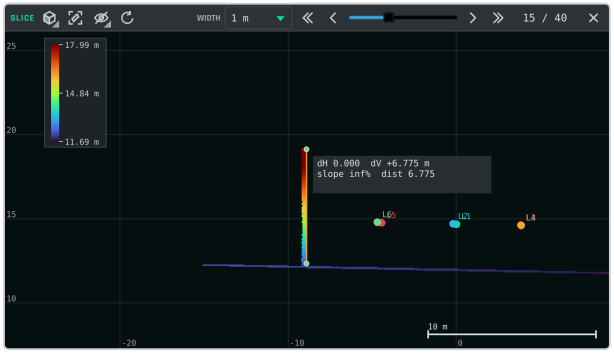


Figure 8.4 Measuring the height of a pole in the Slice.

With the i tool (`I`), click a point: its coordinates, time, line, file, laser, return, range, azimuth, intensity, class and status appear in the Info tab (Lines · Classes · Info). A crosshair marks the point under the mouse.

Undo, rejected points

- Every edit is written to the point store at once and listed in the History tab. Undo (`Ctrl+Z`) and redo (`Ctrl+Y`) have no fixed number of steps; up to 60 million changed points are kept, after which the oldest steps are dropped.
- Every point is either **accepted** or **rejected**. Rejected points are never deleted: they stay in the point store and are only hidden, so rejecting points by hand or with a filter makes them disappear from the views (unless that view's eye shows rejected points) and they can be brought back at any time. Views draw and colour only the points they show, so hiding the rejected points also makes them faster. Filters work on accepted points only, and rejected points are not used as neighbours. Each rejected point keeps the reason it was rejected: *manual*, *automatic* (a filter) or *original*. The filter *Reset modifications* re-accepts by reason, so it can, for example, take back only the automatic rejections.
- Making an edit while a filter preview is shown ends the preview without accepting it.

9 Automatic filters

Filters are kept in **groups**. A group runs its filters in order, each on the result of the one before. Nothing changes until you accept the preview or run the group.



Figure 9.1 The Automatic filters tab with the group *Feature extraction* and its sixth filter, *Poles (voxel layers)*.

ROW	CONTROLS
`GROUP`	The group list; `?` explains the group and lists its filters; the menu button ☰ (below).
`FILTER`	The switch turns the filter on or off (off filters are skipped and marked *(off)*); the list of the group's filters, numbered in running order; `?` gives what the filter does, every setting and the source paper; `+` adds a filter after this one, `–` removes it, the copy button duplicates it, the arrows run it earlier or later.
Settings	The selected filter's settings fill the space below. Hover a setting for its help. Every filter ends with `ONLY CLASSES` and `ONLY LINES` (comma lists such as `1,2` or `1,3-4`; empty = all).
Bottom	`LIVE PREVIEW`, `PAUSE`, `ACCEPT`, `SCOPE`, `RUN`, and the result of the last preview or run.

Preview, accept, run

1. Choose a group, and the `SCOPE`: *Inspection Area*, *Slice*, *Selected lines (whole project)* or *Entire project*.
2. Press `LIVE PREVIEW`. The group runs in the background on the loaded points (on the slice only with the scope *Slice*) and the views show the result: orange points will be rejected, green ones re-accepted, and the Classes list and Classification colours show the previewed classes. The result line gives the counts, the time of each filter and their messages.
3. Change settings: the preview updates by itself shortly after each change. Press `PAUSE` while you change several, and again to update.
4. Press `ACCEPT` to commit the previewed changes to the loaded points (one undo step), or `LIVE PREVIEW` again to drop the preview.
5. Or press `RUN` to apply the group to the scope directly.

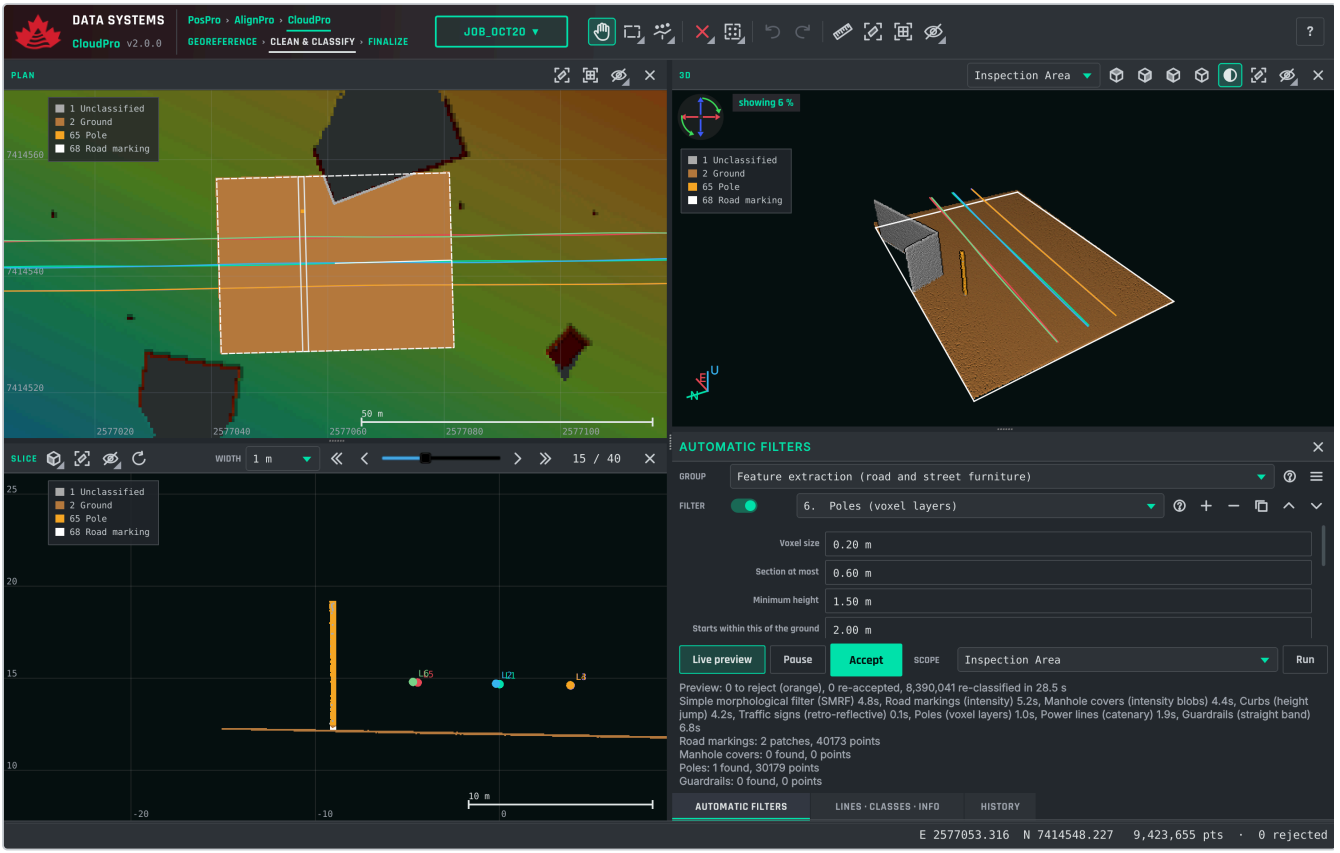


Figure 9.2 Live preview of the group *Feature extraction*: Plan, Slice and 3D coloured by classification show the poles (65) and road markings (68) found; the result line lists the time of each filter and what it found.

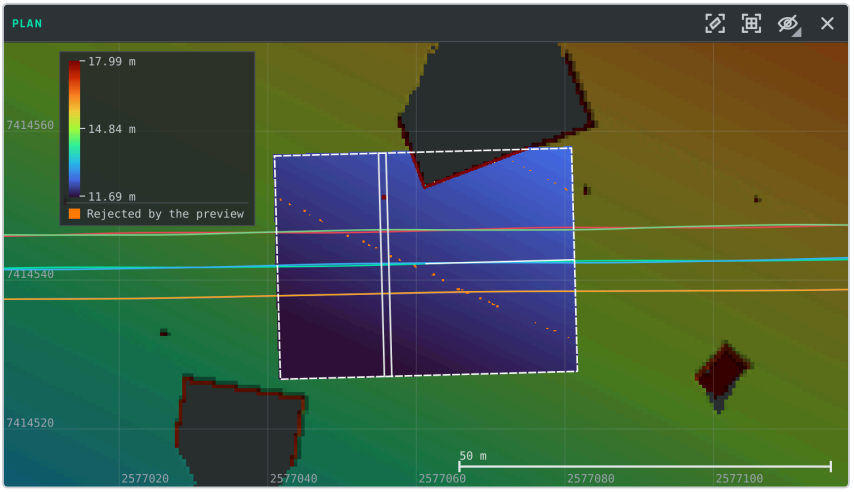


Figure 9.3 Preview of the group *Snow and rain*: orange points would be rejected; green ones, rejected by an earlier run, would be re-accepted by its first filter, *Reset modifications*.

SCOPE	RUN
Inspection Area, Slice	Runs in the background on the loaded points; one undo step.

Selected lines (whole project)	Asks first, then edits the stored points part by part: 50 m tiles with a 10 m buffer, so neighbourhood filters see across tile edges; a tile with more than 2 million points is split into smaller squares, and where the points are so dense that a part with its buffer would hold more than 6 million, the buffer shrinks (not below 3 m). Parts run at the same time on the processor's cores, as many as fit in half the free memory. The progress line shows the parts done, the points done, how many parts are running, the points changed and the time left. `CANCEL` stops within seconds and writes nothing; the stored points stay as they were. *Selected lines* are the line chosen in the tool bar's line list, else the lines ticked in the Lines list. The run is one undo step when it changes up to 20 million points; above that it cannot be undone (on projects of 20 million points or more the question says so): *Reset modifications* then re-accepts automatic rejections.
Entire project	

Every accepted preview and run is recorded with its group, settings, scope and number of points changed; Finalize writes these into the report and `cloud.json`.

Groups

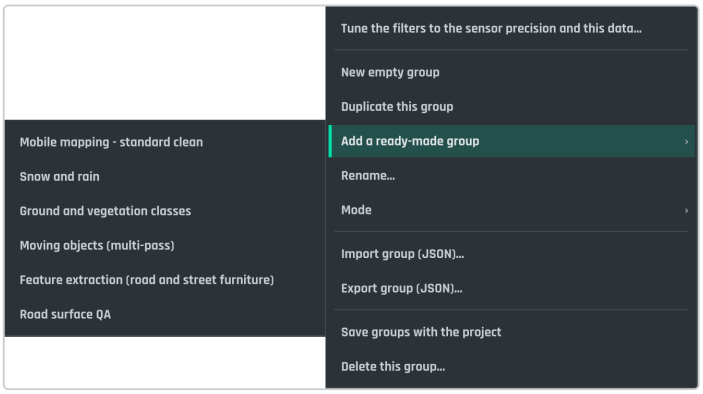


Figure 9.4 The group menu (☰) with the ready-made groups.

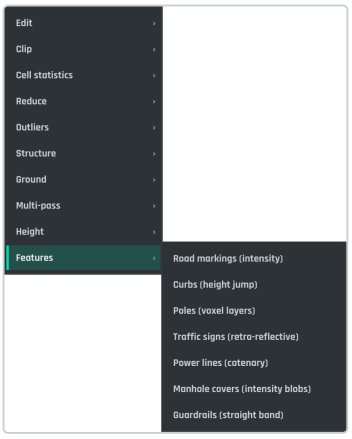


Figure 9.5 `+` lists the filters by kind; hover one for what it does.

ITEM	DOES
Tune the filters to the sensor precision and this data...	See Tuning.
New empty group, Duplicate this group	Adds a group (the copy is named *(copy)*).
Add a ready-made group	Adds a fresh copy of one of the groups below.
Rename...	Renames the group.
Mode	*Combined*: all lines filtered together (lines that fit well). *Individually*: each line filtered on its own (lines that do not fit in height). *Reset modifications* and *Edit classification* always run combined.
Import group (JSON)..., Export group (JSON)...	Share groups between projects.
Save groups with the project	Writes `filters/groups.json` (Save and the 2-minute autosave do the same).
Delete this group...	After a confirmation; the last group cannot be deleted.

Ready-made groups added in a later version are added to older projects' lists automatically.

READY-MADE GROUP	FILTERS, IN ORDER
Mobile mapping - standard clean	Reset modifications · Vehicle self-hits · Isolated cluster (0.5 m, fewer than 8 points) · Statistical outlier (k 12, SD multiplier 3) · Snow / rain / dust (DIOR) · SMRF · Height above ground (reject more than 0.25 m below ground)

Snow and rain	Reset modifications · Snow / rain / dust (DIOR, intensity threshold 20) · Isolated cluster (0.4 m, fewer than 5 points) · Low hit count (0.4 m, fewer than 3)
Ground and vegetation classes	Inverted cloth (0.5 m, clip 0.2 m) · Height above ground (classify vegetation) · Vertical plane (classify walls, nothing behind)
Moving objects (multi-pass)	Moving objects / ghosts · Isolated cluster (fewer than 20 points)
Feature extraction (road and street furniture)	SMRF · Road markings · Manhole covers · Curbs · Traffic signs · Poles · Power lines · Guardrails
Road surface QA	Surface fit (class 2, gate 0.05 m) · 95% confidence level (class 2, limit 0.04 m)

Adding filters and their help

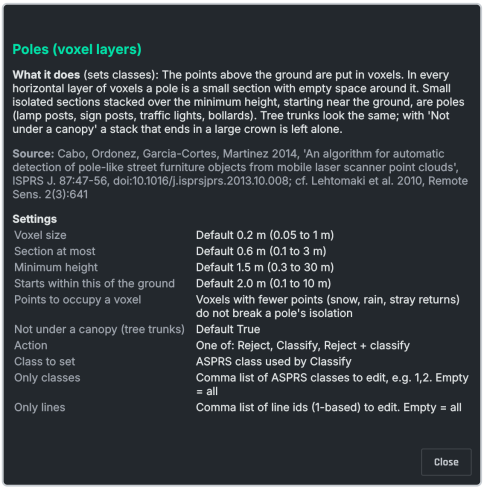


Figure 9.6 The `?` of a filter: what it does, its source and every setting.

Filters that need a reference raster (*Clip reference surface*) show a `REFERENCE RASTER` list above their settings: the raster layers of the QGIS project, or `CHOOSE A RASTER FILE…`.

Tuning to the sensor precision

☰ › `TUNE THE FILTERS TO THE SENSOR PRECISION AND THIS DATA…` measures the noise of the loaded Inspection Area and combines it with the error budget of the hardware: the Hesai XT32 (range precision 0.5 cm, 1 sigma) and the Certus Evo, or, when PosPro finalized one, the standard deviations of the trajectory actually used (SMRMSG). Load an Inspection Area on a road seen by two or more passes first; with fewer than 1,000 points loaded CloudPro asks you to.

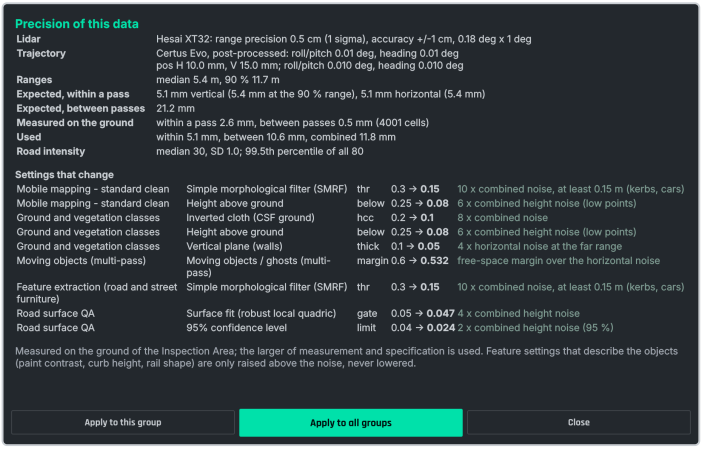


Figure 9.7 Tuning: the precision of this data, and every setting that would change, old → new, with the reason.

The noise within a pass is measured on the ground of each line, and between passes on the ground seen by two lines; the larger of measurement and specification is used. Height and distance tolerances follow the noise. Feature settings that describe the objects (paint contrast, curb height, rail shape) are only ever raised above the noise, never lowered. Choose `APPLY TO THIS GROUP`, `APPLY TO ALL GROUPS` or `CLOSE`. The tuning is written to a log and the measured precision is kept in the project.

The 36 filters

Default values are in brackets. The action of the feature, eigenvalue and moving-object filters is *Reject*, *Classify* or *Reject + classify*, with `CLASS TO SET`. The source of each finalized method is given by its `?` and in `docs/REFERENCES.md` in the plugin folder.

EDIT

FILTER	WHAT IT DOES	MAIN SETTINGS
Reset modifications	Re-accepts rejected points by reason, removes height shifts and/or resets the classification.	Re-accept automatic (on), manual (off), original (off) rejections; Remove height shifts; Reset classification to 1.
Edit classification	Sets the class and/or status of the points that meet a condition. Put it last with *Rejected by this group* to mark which group rejected a point.	Only when point: Any, Is accepted, Is rejected, Rejected - automatic / manual / original, Rejected by this group, Modified by this group; Set class (−1 = keep); Set status: Keep, Accept, Reject.

CLIP

FILTER	WHAT IT DOES	MAIN SETTINGS
Clip absolute height	Rejects points outside a height range.	Minimum (−100 m), Maximum (1000 m); Clip outside / Clip inside.
Clip intensity	Absolute: rejects outside a range. Relative: rejects outside the cell mean ± k·SD of the cell and its neighbour rings (high, low or both).	Mode; Minimum (0), Maximum (255); Cell size (1 m); Neighbour rings (1); SD multiplier (3); Range-normalise.
Clip range / sensor geometry	Rejects returns by range, height above or below the sensor, side of the vehicle, azimuth window.	Min. (0) / max. range (200 m); Max. height above (100 m) / depth below sensor (100 m); Side: Both, Port only, Starboard only; Azimuth window (0–360°).
Clip laser channel / return	Rejects by laser id and/or return type.	Reject laser ids (e.g. `1,2,30-32`); Reject return: None, Last (1st block), Strongest (2nd block).
Clip time	*Newest per cell* keeps, per cell, only points newer than the newest minus the maximum age (the latest pass over a changed area); *Window* rejects outside a UTC time of day.	Mode; Max. age (60 s); Cell size (1 m); Window from / to (s of day UTC).
Vehicle self-hits	Rejects returns in a box around the Lidar in the vehicle frame (roof rack, mast, the vehicle, people around a parked vehicle).	Forward (2.5 m), Backward (3 m), Half width (1.2 m), Above (0.5 m), Below sensor (1.6 m; keep it above the road).
Clip reference surface	Rejects points outside a gate around a reference raster (DEM, earlier survey, design). Cells without reference are ignored.	Reference raster; Max. above (0.3 m) / below reference (0.3 m).

CELL STATISTICS

FILTER	WHAT IT DOES	MAIN SETTINGS
95% confidence level	Per cell, rejects the point furthest from the median until 1.96·SD is below the limit or the minimum count is reached. For surfaces (use Only classes = 2 after ground classification).	Cell size (0.5 m); Max. 95% confidence (0.05 m); Min. points in cell (5); Max. iterations (50).
Low hit count	Rejects all points of a cell or voxel holding fewer accepted points than the minimum.	Cell size (0.5 m); Min. points (3); 2-D cells / 3-D voxels.
Clip mean height	Rejects points further than the gate from the mean of the cell and its neighbour rings; *Point centred* uses a cell around each point, without it.	Cell size (1 m); Reject outside mean ± (0.3 m); Neighbour rings (1); Point centred.
Lonely cells	Rejects points in cells not connected by a side to the main survey cells.	Cell size (0.5 m); Keep groups of at least (cells; 0 = only the largest).

REDUCE

FILTER	WHAT IT DOES	MAIN SETTINGS
Reducer (voxel)	Keeps the point closest to each voxel centre and rejects the others; with a few mm it removes duplicate points.	Horizontal (0.05 m) and vertical voxel size (0.05 m).
Reducer (scan based)	Keeps every Nth rotation of the scanner and/or every Nth laser channel.	Keep every Nth rotation (1); Keep every Nth channel (1).

OUTLIERS

FILTER	WHAT IT DOES	MAIN SETTINGS
Statistical outlier (SOR)	Rejects points whose mean distance to their k nearest neighbours is above the mean + α·SD of all points; *Range adaptive* scales the distances by range first. Run it early (Rusu et al. 2008).	Neighbours k (12); SD multiplier (2.5); Range adaptive (on).
Radius outlier (ROR)	Rejects points with fewer than N neighbours within radius R.	Radius (0.3 m); Min. neighbours (3).
Snow / rain / dust (DROR, LIOR, DIOR)	DROR: radius outlier removal with a radius growing with range. LIOR / DIOR first keep every bright return and test only the dim ones, which protects signs, markings and thin structures (Charron et al. 2018; Park et al. 2020; Roriz et al. 2022).	Method (DIOR); Beta (3); Azimuth step α (0.18°: the XT-32 at 600 rpm; × N when keeping every Nth packet); Min. search radius (0.04 m); Min. neighbours (3); ROR radius (0.2 m); Intensity threshold (18); Only within range (40 m).
Isolated cluster	Groups points within the point interval of each other and rejects small clusters, or all but the largest. Z scale < 1 makes it less sensitive vertically.	Point interval (0.5 m); Reject clusters smaller than (10 points); Mode; Z scale (1).
Slope / spike filter	Rejects points steeper than the limit to their neighbours that stand out above or below all of them. For surfaces (Vosselman 2000).	Neighbours (8); Max. mean slope (1); Direction: Both, Spikes up, Spikes down; Min. height difference (0.05 m).
Surface fit (robust local quadric)	Fits a robust plane or quadric through each point's neighbours (without the point) and rejects it when further than the gate. Run it after the large spikes are gone.	Neighbours (16); Reject gate (0.08 m); Surface: Plane, Quadric.

STRUCTURE

FILTER	WHAT IT DOES	MAIN SETTINGS

Vertical plane (walls)	Finds vertical planes (facades, walls) by RANSAC. Points in the plane can be classified (6 building); points behind it, seen from the Lidar, are usually reflections through glass or interior returns and can be rejected.	Thickness (0.1 m); Max. tilt (5°); Min. points per plane (2000); Min. wall height (1 m); Max. planes (6); Inside: Nothing / Classify / Reject, class (6); Behind: Nothing / Reject / Classify, class (7), further than (0.3 m).
Geometric features (eigenvalues)	From the local covariance: scattered (noise, vegetation), linear (wires, poles, edges), planar vertical (walls), planar horizontal (ground, roofs) (Demantké et al. 2011; Weinmann et al. 2015).	Select (Scattered); Neighbours k (16); Feature threshold (0.5); Verticality threshold (0.85); Action (Classify), class (1).

GROUND

FILTER	WHAT IT DOES	MAIN SETTINGS
Inverted cloth (CSF ground)	Turns the cloud upside down and drops a simulated cloth on it; points near the settled cloth are ground (Zhang W. et al. 2016).	Cloth resolution (0.5 m); Scene: 1 Steep slope, 2 Relief, 3 Flat, 4 Super flat; Max. iterations (500); Clip distance (0.25 m); Slope smoothing; Class for non-ground (1; −1 = keep); Reject non-ground.
Simple morphological filter (SMRF)	Progressive opening of a minimum surface, then an elevation threshold growing with the slope. Few settings, robust in towns (Pingel et al. 2013).	Cell size (0.5 m); Slope (0.15); Max. window (18 m); Elevation threshold (0.3 m); Elevation scalar (1.25); Class for non-ground; Reject non-ground.
Progressive morphological filter (PMF)	Openings of a minimum grid with a growing window and a slope-dependent height threshold (Zhang K. et al. 2003).	Cell size (0.5 m); Slope s (1); Max. window (20 m); Initial (0.15 m) and max. height threshold (2.5 m); Window growth; Final ground tolerance (0.25 m); Class for non-ground; Reject non-ground.
Height above ground	Builds a DEM from the ground class (2) and rejects points below the ground (multipath, low noise) or high above it, or classifies vegetation 3 / 4 / 5 or a height band among the unclassified points. Needs a ground filter first.	DEM cell (0.5 m); Mode: Reject below ground, Reject above, Classify vegetation, Classify band; Below by more than (0.25 m); Above by more than (40 m); Low (0.5 m) / medium vegetation up to (2 m); Band from / to, class.

MULTI-PASS AND HEIGHT

FILTER	WHAT IT DOES	MAIN SETTINGS
Moving objects / ghosts (multi-pass)	Follows the laser rays of each pass through a voxel grid. A point is a moving object (car, pedestrian, cyclist) when another pass saw through its voxel. Needs two or more lines over the same place (Schauer & Nüchter 2018).	Voxel size (0.25 m); Min. free-space rays (2); Ray sampling (1 in 3); Max. ray length (40 m); Stop before the return (0.6 m); Protect voxels next to a pass's own returns (1); Action (Reject), class (64).
Shift height	Shifts the points up (+) or down (−), added to any existing shift.	Shift (0 m).

FEATURES

The feature filters need ground first: the ready-made group starts with SMRF. Their intensity thresholds were set on XT-32 data; check them on your first job, or run Tuning.

FILTER (CLASS)	WHAT IT DOES	MAIN SETTINGS

Road markings (intensity) (68)	Road points near the drive line are split into bands by distance, each with its own Otsu threshold; bright points are gridded, gaps closed, small patches dropped (Guan et al. 2014).	Image cell (0.1 m); Band width (2.5 m); Minimum contrast (40); Road up to (0.15 m) above ground; Only ground-class points; Minimum area (0.05 m²); Distance from the drive line (0–15 m).
Curbs (height jump) (69)	Cells whose neighbourhood rises by a curb's height over a short distance with a steep face, in runs of a minimum length (Yang et al. 2013).	Grid cell (0.1 m); Window (0.3 m); Height jump 0.08–0.3 m; Steeper than (40°); Minimum length (2 m); Distance from the drive line (1–20 m).
Poles (voxel layers) (65)	Small isolated sections stacked over the minimum height, starting near the ground. With *Not under a canopy* tree trunks are left alone (Cabo et al. 2014).	Voxel size (0.2 m); Section at most (0.6 m); Minimum height (1.5 m); Starts within (2 m) of the ground; Points to occupy a voxel (3); Not under a canopy (on).
Traffic signs (retro-reflective) (67)	Bright clusters at sign height that are planar, upright and of sign size; the whole plate is then classified (Riveiro et al. 2016).	Intensity at least (180); Height above ground 1.2–7 m; Cluster distance (0.3 m); Minimum points (15); Planarity (0.7); Area 0.1–4 m²; Include the plate within (0.06 m).
Power lines (catenary) (14)	High points with free space above and below, clustered along their length, kept when long, straight in plan and fitting a catenary (Guan et al. 2016).	Higher than (4 m); Voxel size (0.25 m); Free space (1 m); Points per voxel (3); Cluster distance (1.5 m); Minimum length (5 m); Fit RMS (0.15 m); Straight within (0.5 m).
Manhole covers (intensity blobs) (70)	Round blobs of the road's intensity image that differ from the background, of a cover's size (Guan et al. 2014).	Image cell (0.08 m); Contrast (20); Covers appear: Brighter, Darker, Either; Diameter 0.4–1.2 m; Roundness (0.55); Road up to (0.1 m); Distance from the drive line (0–10 m).
Guardrails (straight band) (71)	A long, straight band of points at rail height beside the road, parallel to the drive line, with nothing above it (Gao et al. 2022).	Height above ground 0.4–1.2 m; Cluster distance (0.5 m); Minimum length (4 m); Line fit RMS (0.1 m); Occupied slices (60%); Parallel within (20°); Points above at most (15%); Distance (0.5–15 m).

10 Lines · Classes · Info and History

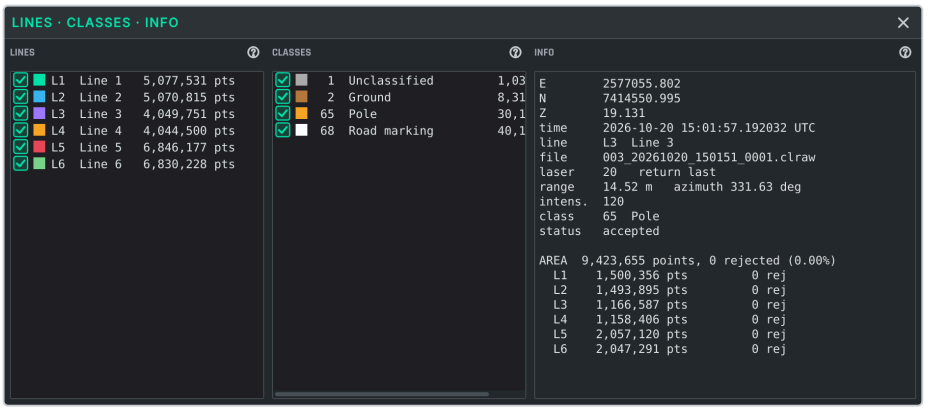


Figure 10.1 The Lines · Classes · Info tab after clicking a pole point with the i tool (the right-hand windows widened by dragging the line between them).

COLUMN	SHOWS AND DOES
LINES	One line per mapper recording, with its colour and point count. Untick to hide its points (hidden points are not edited). Right-click a line: SHOW ONLY THIS LINE, SHOW ALL LINES, EDIT ONLY THIS LINE (sets the tool bar's line list), INSPECTION AREA FOLLOWS THIS LINE.
CLASSES	The classes in the loaded Inspection Area with their colour and count (during a preview, the previewed classes). Untick to hide a class.
INFO	The point clicked with the i tool or double-clicked in 3D: E, N, Z, time (UTC, to the microsecond), line, file, laser and return, range and azimuth, intensity, class, status (accepted, or rejected and why) and any height shift. Below, the Inspection Area: points and rejections, the selection, and the points and rejections per line.

The ? of each column explains it.

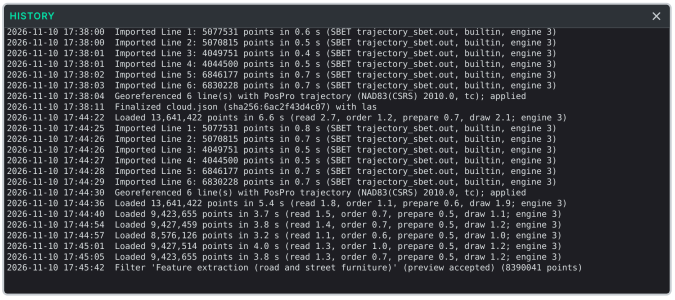


Figure 10.2 The History tab: imports, georeferencing, finalizing, filter runs and edits with their UTC time.

History lists the last 400 entries of the project's history: imports and georeferencing, filter runs and accepted previews, manual edits with their number of points, undo and redo, and finalizing.

11 Project Properties

PROJECT MENU › PROJECT PROPERTIES… or `Ctrl+R` (also in the project box) opens a window that can stay open, or be minimised, while you work. Nothing in it changes the raw captures.

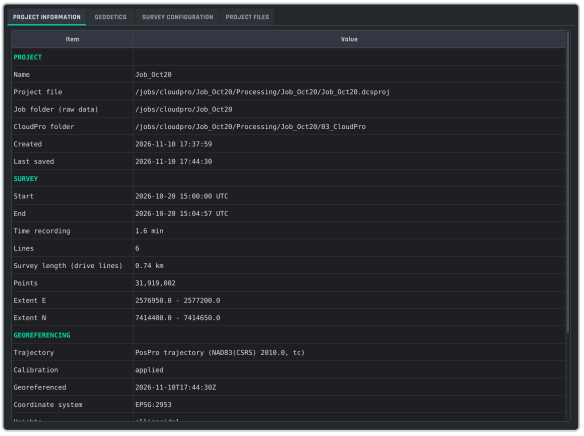


Figure 11.1 Project information.

Project information lists the project (name, project file, job folder, CloudPro folder, created, last saved), the survey (start, end, time recording, lines, survey length of the drive lines, points, extent, location of the centre in latitude and longitude), the georeferencing (trajectory, calibration, when, coordinate system, heights), the chain with the state of each stage, and the last publication.

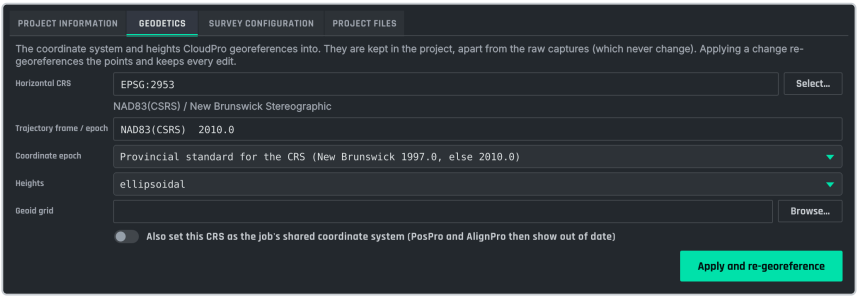


Figure 11.2 Geodetics.

Geodetics sets the coordinate system and heights CloudPro georeferences into: **HORIZONTAL CRS** (type an EPSG code or **SELECT…**), the trajectory frame and epoch (read only), **HEIGHTS** ellipsoidal or orthometric with a **GEOID GRID**, and **ALSO SET THIS CRS AS THE JOB'S SHARED COORDINATE SYSTEM** (PosPro and AlignPro then show out of date). **APPLY AND RE-GEOREFERENCE** opens the Georeference dialog, keeping every edit; the new CRS and heights (and the shared CRS, when ticked) take effect only when the georeferencing has finished, so cancelling leaves the project as it was. For orthometric heights a geoid grid is required.

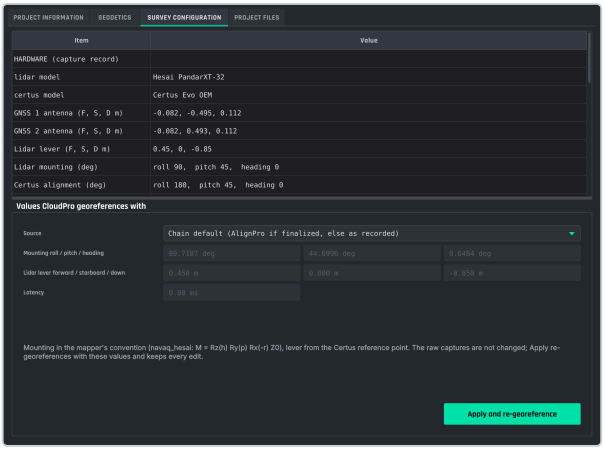


Figure 11.3 Survey configuration.

Survey configuration shows the hardware recorded in the captures (Lidar and Certus models, GNSS antennas, Lidar lever arm and mounting, Certus alignment, coordinate system, vehicle, connections), PosPro's trajectory (frame, mode, fixed %, maximum sigmas, base station, levers, id) and AlignPro's calibration (mounting, lever arm, latency, boresight, average error, id). Below it, **Values CloudPro georeferences with**:

SOURCE	USES	RESULTS SAY
Chain default (AlignPro if finalized, else as recorded)	AlignPro's calibration, else the recorded mounting	*applied* or *no calibration*
Recorded in the capture (no calibration)	The mounting recorded in the capture, even when AlignPro finalized	*no calibration*
Entered here	The mounting roll / pitch / heading, Lidar lever forward / starboard / down and latency typed here	*entered values*

The mounting is in the mapper's convention, the lever arm from the Certus reference point. `APPLY AND RE-GEOREFERENCE` keeps the values in the project and opens the Georeference dialog; if you cancel it, the points are shown as out of date until you update.

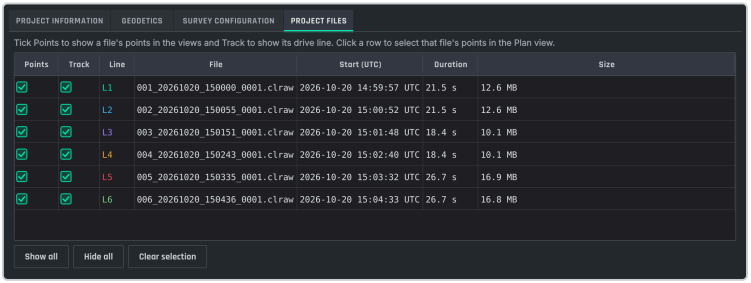


Figure 11.4 Project files.

Project files has one row per `.clraw` file: `POINTS` shows or hides its points in the views, `TRACK` its drive line in the Plan; line, file, start (UTC), duration and size. Click a row to select that file's points in the Inspection Area and draw its drive line thicker (move the area onto the drive line if no points are selected). `SHOW ALL`, `HIDE ALL`, `CLEAR SELECTION`.

12 Finalizing, exporting and QGIS layers

Finalize is the job's delivery, in `03_CloudPro/out/`, recorded in the chain. **Export** writes a copy anywhere, for any purpose.

Finalize

PROJECT MENU › FINALIZE FOR THE JOB (LAS, GROUND GRID, CLOUD.JSON)… (`Ctrl+P`). The project must be a job project of the chain and georeferenced.

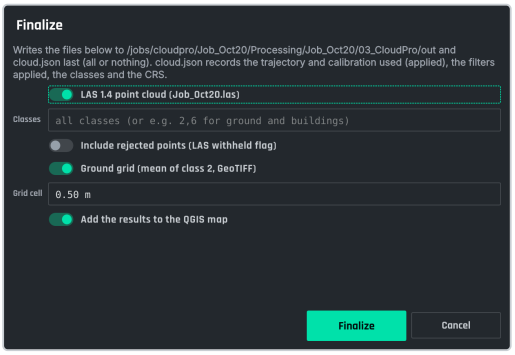


Figure 12.1 The Finalize dialog with the ground grid ticked. Out-of-date reasons and *no calibration* would be listed in amber above the options.

OPTION	EFFECT
LAS 1.4 POINT CLOUD (<NAME>.LAS)	The whole cloud, all lines.
CLASSES	Empty for all classes, or a list (e.g. `2,6`).
INCLUDE REJECTED POINTS (LAS WITHHELD FLAG)	Off: rejected points are left out. On: they are written with the withheld flag.
GROUND GRID (MEAN OF CLASS 2, GEOTIFF), **GRID CELL**	A grid of the mean height of the accepted ground points (gaps of up to 2 cells filled), at the chosen cell (0.5 m).
ADD THE RESULTS TO THE QGIS MAP	Adds the LAS and the grid to the CloudPro group of the QGIS project.

At least one of the LAS and the ground grid must be ticked. Finalizing runs in the background and is all or nothing: the files are written under temporary names and renamed at the end.

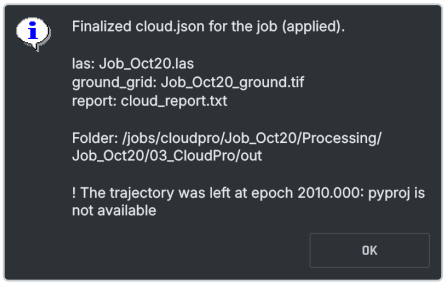


Figure 12.2 The result: the calibration state, the files and the folder.

FILE IN OUT/	CONTENT
`<Name>.las`	LAS 1.4, point data record format 6, with the CRS as WKT. User data holds the laser channel (0–31, the laser id minus one), point source id the line number, GPS time in adjusted standard time; the scan angle is 0.
`<Name>_ground.tif`	The ground grid (an ESRI ASCII `.asc` grid when GDAL is not available).

`cloud_report.txt`	Readable report: inputs (trajectory, calibration, mounting, lever arm, latency, CRS, heights), lines, point and class counts, filters applied with their settings, outputs, warnings.
`cloud.json`	Written last, through DCS Core: the calibration state, the trajectory and alignment results the points were really georeferenced with, the georeferencing values, CRS (frame, epoch, heights, geoid), lines and files, point and class counts (accepted, rejected manual and automatic), every filter run with its settings, the outputs and the warnings.

Because `cloud.json` names the PosPro and AlignPro results used, CloudPro shows ⚠ **stale** when they finalize again, until you update the georeferencing and finalize again.

Export

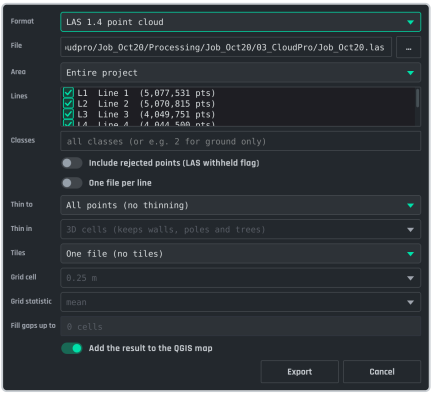


Figure 12.3 The Export dialog.

SETTING	CHOICES
FORMAT	*LAS 1.4 point cloud* (as finalized), *XYZ text (CSV)* with the columns E, N, Z, I (intensity), C (class), L (line), T (UTC time), or *Grid surface (GeoTIFF)*.
FILE	Defaults to `03_CloudPro/<Name>.las`; the extension follows the format.
AREA	*Entire project* or *Inspection Area*.
LINES, **CLASSES**	Ticked lines; empty for all classes, or a list.
INCLUDE REJECTED POINTS, **ONE FILE PER LINE**	LAS only. One file per line adds `_L001`, `_L002`… to the name.
THIN TO, **THIN IN**	LAS and XYZ. Choose a grid spacing from the list (0.02 to 5 m; the point density it gives is shown beside it, e.g. *0.1 m (100 pts/m²)*), or *All points*. One real point is kept per cell, the one nearest the cell centre, so coordinates (3 decimals) and attributes are untouched. *3D cells* keeps walls, poles and trees; *2D plan cells* keeps one point per plan cell, for ground models. Cells line up with the project grid, so tiles and lines thin consistently.
TILES	LAS and XYZ. *One file*, or 50 to 1000 m tiles: each file is named by its tile's lower-left corner in whole metres, e.g. `Shirley_231400_5080900.las` (with one file per line: `Shirley_L001_231400_5080900.las`). Tiles without points make no file. Above 400 tiles CloudPro asks first; at most the first 50 are added to the QGIS map.
GRID CELL, **GRID STATISTIC**, **FILL GAPS UP TO**	Grid only: cell (0.25 m); mean, min, max, count or intensity; empty cells filled from the nearest value up to N cells away. Grids use accepted points only.
ADD THE RESULT TO THE QGIS MAP	Adds the file to the CloudPro group of the QGIS project.

QGIS layers

- **PROJECT MENU › PICK THE INSPECTION AREA ON THE QGIS MAP** : the QGIS message bar says *Click on the map to centre the Inspection Area there*; the click is converted to the project's coordinate system and the area moves there.
- **PROJECT MENU › ADD THE DRIVE LINES AND INSPECTION AREA TO QGIS** : adds two temporary layers in a **CloudPro** group: *<Name> - drive lines* (with line number, name and points) and *<Name> - inspection area*, and zooms the map to them.
- Finalize and Export add their LAS (as a point cloud layer) and grids to the same group when *Add … to the QGIS map* is ticked.
- Raster layers of the QGIS project are offered as reference surfaces in *Clip reference surface*.
- **SAVE THE QGIS PROJECT IN THE PROJECT FOLDER** (project box) writes `<Name>.qgz` beside the `.dcsproj`.

13 The command-line tool

The same steps run without the window, with the same project, inputs, entered values and filter groups. Run them with the QGIS Python (the OSGeo4W shell on Windows) from the folder that holds the `cloudpro` and `dcs_core` plugin folders, e.g. `%APPDATA%/QGIS/QGIS3/profiles/default/python/plugins`.

```
python -m cloudpro.cli import  D:/Job/Processing/Street/Street.dcsproj
python -m cloudpro.cli update  D:/Job/Processing/Street/Street.dcsproj
python -m cloudpro.cli filter  D:/Job/Processing/Street/Street.dcsproj --group "Mobile mapping - standard clean"
python -m cloudpro.cli finalize D:/Job/Processing/Street/Street.dcsproj --dem 0.5
python -m cloudpro.cli info    D:/Job/Processing/Street/Street.dcsproj
```

COMMAND	DOES	OPTIONS
`import`	Georeferences the job's captures with PosPro's trajectory and AlignPro's mounting (or the fallbacks), carrying edits over. Prints the inputs and warnings.	`--clraw` files or folders instead of the job's captures; `--returns both|last|strongest`; `--rmin`, `--rmax` (1, 120 m); `--mount R P H` and `--lever F S D` override the mounting (then *no calibration*).
`update`	Re-georeferences with the current inputs, keeping every edit.	–
`filter`	Runs a group on the whole project, tile by tile, and records it.	`--group` name (the project's groups, then the ready-made ones); `--lines` e.g. `1,3-4`.
`publish`	Finalizes LAS, optional ground grid, report and `cloud.json`.	`--dem` cell (0 = no grid); `--classes`; `--include-rejected`.
`info`	Name, CRS, lines and points, coverage (area with points, median density, share at or above the target), the state of each stage and what is out of date.	`--target 100` (pts/m²), `--cell 1` (m)
`export`	Writes LAS, XYZ or a grid.	`--las`, `--xyz`, `--grid` file; `--cell` (0.5 m), `--stat mean|min|max|count|intensity`, `--classes`, `--lines`, `--per-line`, `--include-rejected`.
`create`	Creates an older-style stand-alone project (no chain).	`folder name --crs EPSG:2953 --tile 50`; for its `import`: `--sbet` (omit for the real-time track), `--bore dR dP dH`.

A run in a job project holds the stage lock and writes a log in `03_CloudPro/log/`.

14 Troubleshooting

MESSAGE OR SYMPTOM	WHAT TO DO
CloudPro needs the DCS Core plugin… (QGIS message bar)	Install DCS Core (chapter 2), then restart QGIS.
No .clraw files were found in the job's captures	Check the job folder, use **PROJECT MENU › ADD OTHER .CLRAW FILES TO THE JOB…**, or relocate a moved job folder.
Raw data not found when opening	Answer Yes and point to the job folder's new location, or later use **RAW DATA MOVED? RELOCATE THE JOB FOLDER…**.
This job's point cloud is open elsewhere	Another QGIS has the job's cloud open for editing. Close it there first.
PosPro has not finalized a trajectory for this project: the Certus real-time track is used	Run PosPro, then the **GEOREFERENCE** step in the header.
The Georeference step's tooltip says *no calibration*	AlignPro has not finalized (or the source is set to the recorded mounting). Run AlignPro, then the **GEOREFERENCE** step in the header.
AlignPro used the … convention, CloudPro expects navaq_hesai: calibration not used	Finalize the calibration again with a current AlignPro.
The AlignPro calibration was solved on an earlier PosPro trajectory	Run AlignPro again on PosPro's current trajectory.
Out of date: …	Press the **GEOREFERENCE** step in the header, then finalize again (Update georeferencing).
the trajectory (…) does not cover line …	The trajectory does not span that capture; process the whole recording in PosPro.
No Zstandard decoder is available to read .clraw files	In the OSGeo4W shell: `python -m pip install zstandard`, then restart QGIS.
.clraw format_version … is newer than this version of DCS Core understands	Update DCS Core and the plugins.
Choose a geoid grid for orthometric heights.	Browse for a geoid grid in Geodetics, or choose ellipsoidal heights.
This Inspection Area holds about N million points…	Load it anyway, make the area smaller (Inspection Area properties) or raise **PROJECT MENU › MEMORY LIMIT**.
3D shows *showing N %*	Normal on large areas; selecting and editing still use every point.
Another task is still running.	Wait for the progress bar to finish, or press **CANCEL**.
Load an Inspection Area with points… (Tuning)	Load an area with at least 1,000 points, ideally a road seen by two passes.
Filter messages: *no ground class yet - run a ground filter first*, *Moving objects: needs two or more lines in the area*, *Clip reference surface: no reference raster selected*, *Vehicle self-hits: no sensor track for these lines*	Add a ground filter (SMRF) before; place the area where two passes overlap; choose a reference raster; georeference the line again.
Features missed or found too often	Run Tuning, then adjust the intensity settings of the filter (its **?** explains each one).
Finalizing is for job projects of the DCS chain…	An older stand-alone project: use Export.
The user guide is not in this installation	Reinstall CloudPro from its zip.

Each georeferencing, project-wide filter run, tuning and publication writes a log in `03_CloudPro/log/`; a failed task shows its error in a message and leaves the earlier results in place.

15 Reference

Keys and mouse

KEY / MOUSE	DOES
`Space`, `G`, `P`, `L`, `B`	Pan, rectangle, polygon, lasso, brush
`U`, `J`, `M`, `I`	Above a line, below a line, measure, point information
`X`, `V`, `S`, `K`, `F`	Reject, accept, select, classify, filter
`H`	Show / hide rejected points in every view
`Delete`, `Ctrl+D`	Reject the selection, clear the selection
`Ctrl+Z`, `Ctrl+Y`	Undo, redo
`PgUp` / `PgDn`, `]` / `[`, `.` / `,`	Next / previous slice
`Ctrl+PgUp` / `Ctrl+PgDn`, `}` / `{`	Five slices on / back
`Shift` + arrows, `Shift+PgUp` / `Shift+PgDn`	Step the Inspection Area
Arrows, `+` / `−`, `Home`	2-D views: pan by the step, zoom, zoom to the points
`Ctrl+Home`	Zoom all views to the Inspection Area
Wheel, `Shift` + wheel	Zoom at the cursor; vertical exaggeration in the Slice
Middle drag, double-click	2-D views: pan with any tool, centre here
`Enter`, `Backspace`, `Esc`	Finish a polygon or line, remove its last point, cancel
3D: left drag, `Shift` + left or middle drag, right drag	Orbit, pan, lasso or rectangle edit
3D: wheel, `Ctrl` + wheel, double-click	Zoom, Z scale, new centre and point information
3D (click in it first): `W` `S` `A` `D` `Q` `E`, `1`–`4`, `O`, `0`	Fly, top / along / across / oblique, orthographic, fit
`Ctrl+R`, `Ctrl+I`, `Ctrl+P`, `Ctrl+E`, `Ctrl+S`	Properties, georeference, finalize, export, save
`F1`, `Shift+F1`	Keys and mouse, this guide

Classes

Classes follow ASPRS LAS 1.4, with CloudPro's classes 64–71. Each class has its own colour, clearly different from every other class and from the colours drawn on top of them (magenta rejected points, orange and green preview points, white selected points):

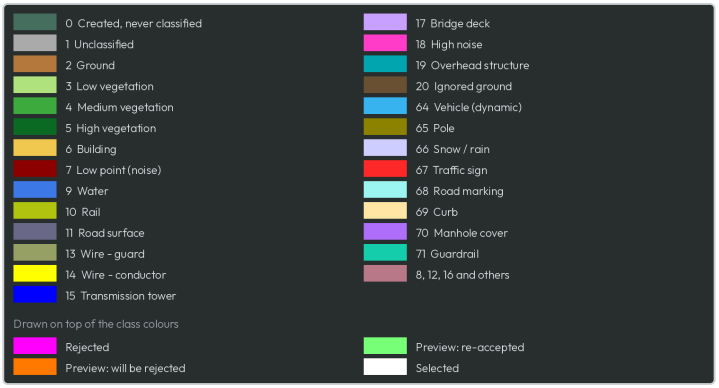


Figure 15.1 The class colours (Colour: Classification).

CLASS	NAME	CLASS	NAME	CLASS	NAME
0	Created, never classified	9	Water	64	Vehicle (dynamic)
1	Unclassified	10, 11	Rail, road surface	65	Pole
2	Ground	13, 14	Wire - guard, wire - conductor	66	Snow / rain
3, 4, 5	Low, medium, high vegetation	15, 17	Transmission tower, bridge deck	67, 68	Traffic sign, road marking
6	Building	18, 19	High noise, overhead structure	69, 70	Curb, manhole cover
7	Low point (noise)	20	Ignored ground	71	Guardrail